

PCR-5D4

Cosmological Population and Transport Extension of the Neutrino Standing-Carrier Space

Public-95 Summary-Target Closure, Source-Faithful Kinetic Saturation, and the Complete-Likelihood Boundary

Exact two-level target scope

The positive theorem is relative to the explicitly frozen summary target $T_{D4}^{\text{pub95-summary}}$: one common standard-thermal physical apex, in-situ retained-oscillation transparency, exact provenance, and membership of the scalar mass-sum coordinate in the released one-sided 95% marginal band. The complete cosmological target T_{D4}^{complete} is a different object. It remains NotReady_{JLR-D4-1} because the fresh correlated joint-DESI evaluator required by JLR-D4-1 is unavailable. The summary theorem is not represented as a complete posterior, a fresh likelihood, or an ordering-conditioned credible interval.

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Abstract

PCR-5B fixes the exact standing-carrier quotient of the neutrino-oscillation target, PCR-5C supplies the functorial target-extension calculus, and SOE-D4-1 authorizes a conservative function-valued source universe of cosmological neutrino histories. This paper gives a two-level PCR-5D4 result that keeps the released scalar summary and the unavailable complete cosmological likelihood as mathematically different targets.

The public release object supplies exact CMB likelihood bytes, the joint DESI archive and covariance records, and a retained scalar mnu marginal whose one-sided weighted 95% quantile is

$$\Sigma_{95}^{\text{pub}} = 0.061934524 \text{ eV}.$$

From that independently frozen record we define the finite summary target $T_{D4}^{\text{pub95-summary}}$. Its empirical predicate is only membership of the scalar mass-sum co-ordinate in

$$B_{95}^{\text{pub}} = (-\infty, \Sigma_{95}^{\text{pub}}].$$

No posterior density, fresh joint-likelihood value, ordering-conditioned probability measure, or complete response geometry is imported into that target.

A source-faithful kinetic-saturation theorem is proved before the universal lift. Starting from the complete target-visible propagation support, it first closes the source trace under every frozen PCR-3/SOE-D4-1 derivation rule and then forms the least finite support closure under every primitive or derived enabled D4 production, collision, decay/injection, hidden or environmental attachment, constituent/collective, and source-to-stress-energy relation. It gives an exact dichotomy: either the closure is exactly the retained propagation support, all off-support kernels vanish by source equations, and the complete D4 neutrino stress-energy diagram descends in a typed finite additive category, once and without double counting, to the three compiled thermal modes; or the history receives its exact fail-closed nontransparent disposition. The typed extension contains no implicit generator from an old-only PCR-3 relation into a D4 kinetic sort. A physically mandatory conversion absent from the frozen derivation theory is readiness- or source-update-bearing rather than a silent same-source consequence.

The theorem is discharged over LAW, LOCAL, HID, COMP, SOL, and HYB by evaluating the actual activity, coupling, threshold, memory, orientation, initial-data, constituent, collective, gluing, and higher-incidence equations. For each PCR-5B base component, SAR-D4-1 first proves that the actual seed-active/off-seed-null Boolean activity vector is a coherent SOE-D4-1 source model by direct inspection of the frozen activity/coherence schemas; STR-D4-1 then applies the kinetic-saturation dichotomy at that already coherent point. No positive conclusion is inferred from the S4-04 all-null expansion; any history with a primitive or derived enabled off-support kinetic or stress-energy incidence remains in the nontransparent branch. The inherited, population, mass-indexed background, cross-incidence, and common-apex residual families then vanish on one semantic physical object. A dedicated background-closure theorem constructs a lawful flat- Λ CDM solution family, compensates the changing neutrino stress energy without leaving the frozen model branch, and distinguishes an untruncated semantic hierarchy from a certified finite truncation. A typed D4-only ideal finally proves in-situ channelwise transparency of the complete retained oscillation packet.

For the frozen rounded oscillation projection,

$$\Sigma_{\text{NO}}(0) = 0.058660254 \text{ eV}, \quad \Sigma_{\text{IO}}(0) = 0.099244289 \text{ eV}.$$

The scalar-band pullback is therefore

$$\Sigma_{\text{NO}}^{-1}(B_{95}^{\text{pub}}) = [0, m_{\star}], \quad \Sigma_{\text{NO}}(m_{\star}) = \Sigma_{95}^{\text{pub}},$$

with

$$m_{\star} \approx 0.002766609601185372 \text{ eV},$$

and

$$\Sigma_{\text{IO}}^{-1}(B_{95}^{\text{pub}}) = \emptyset.$$

These are exact set-theoretic pullbacks of a released scalar marginal band, not transported posteriors.

Consequently, for the summary target,

$$\text{im } \rho_{D4}^{\text{pub95-summary}} = Q_{\text{osc}}^{\text{SCIC}}, \quad |\pi_0 \mathcal{L}_{D4}^{\text{pub95-summary}}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}},$$

so

$$\text{Status}(D4; T_{D4}^{\text{pub95-summary}}) = \text{FullBaseContinuum}_{D4}^{\text{pub95-summary}}.$$

The minimal proved separator on the positive fixed-history branch is Σ_ν , activated by the scalar-valued target evaluator S_Σ . The broader CosmologicalScaleHistory object is retained as a response-ledger class, not promoted into a minimal separator without a variable-history calculation. The compiled source has profile $Y_{\text{compiled},3\text{-block}}$ at the declared macroblock resolution; the fine PCR-5C deletion hypergraph remains explicitly readiness-gated.

The complete cosmological target remains

$$\text{Status}(D4; T_{D4}^{\text{complete}}) = \text{NotReady}_{\text{JLR-D4-1}}.$$

Nonstandard histories remain ResponseOutOfScope, NotReady, or update-bearing according to their exact failure stage. Thus the paper establishes the strongest defensible full-base continuum theorem while preserving the missing fresh likelihood as a named downstream gate rather than turning PCR-5D4 into an executable-formalization project.

Keywords. AASC; neutrino cosmology; neutrino mass; thermal history; free streaming; standing carrier; SCIC; target extension; lift fibre; public marginal summary; DESI; Planck; ACT; mass ordering.

Principal theorem

Let $T_{D4}^{\text{pub95-summary}}$ be the frozen public-95 summary target defined from the exact released scalar band B_{95}^{pub} , together with the source, one-point physicality, retained-transparency, approximation, and provenance requirements stated below. Let

$$\begin{aligned} R_{D4}^{\text{pub95-summary}} &: \mathcal{N}_{D4}^{\text{pub95-summary}} \longrightarrow \mathcal{N}(T_{\text{osc}}), \\ \rho_{D4}^{\text{pub95-summary}} &= \pi_0(R_{D4}^{\text{pub95-summary}}), \\ \mathcal{L}_{D4}^{\text{pub95-summary}}(q) &= \mathcal{N}_{D4}^{\text{pub95-summary}} \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q]. \end{aligned}$$

Then:

1. KSD-D4-1 closes each complete source trace under every source-active D4 kinetic and stress-energy incidence and gives the exact transparent/nontransparent dichotomy;
2. SAR-D4-1 proves a seed-active relative extension functor and the coherence of the compiled partial activity vector; STR-D4-1 then constructs, in every base component, the transparent section on that coherent source stratum; no representative-independent canonical section is assumed;
3. every $q \in Q_{\text{osc}}^{\text{SCIC}}$ consequently has a source-faithful, one-point, retained-transparent standard-thermal normal-ordering lift for every $m_0 \in [0, m_\star]$;
4. a mass-indexed flat- Λ CDM background family closes the Friedmann, conservation, stress-energy, and perturbation equations across that full interval while preserving the frozen present matter anchor;
5. all inherited, pure-D4, cross-incidence, common-cone, conservation, and continuation residuals vanish on the same physical object, and the complete proof applies to LAW, LOCAL, HID, COMP, SOL, and HYB base presentations;
6. unequal m_0 give unequal enriched SCIC classes because the target-fixed scalar evaluator S_Σ is standing-bearing and Σ_{NO} is strictly increasing;

7. the summary-target restriction image and fibre lower bounds are

$$\text{im } \rho_{D4}^{\text{pub95-summary}} = Q_{\text{osc}}^{\text{SCIC}}, \quad |\pi_0 \mathcal{L}_{D4}^{\text{pub95-summary}}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}};$$

8. every standard inverted-ordering point lies outside B_{95}^{pub} and receives status `PublicMarginalBandOutOfBand`, not `SolvedEmpty`;

9. the minimal new separator on the fixed positive branch is Σ_ν ; `CosmologicalScaleHistory` is the broader ledger class; the D1 relation is `ScaleHistoryRefinement`, and the D3 relation is `DistinctSeparator`;

10. the compiled source profile is $Y_{\text{compiled},3\text{-block}}$ at the declared macroblock resolution, the complete source/background/response dependency antichain is recorded separately, and the fine PCR-5C source-deletion profile remains `NotReadyFineSourceDeletion`;

11. the complete cosmological target remains

$$\begin{aligned} \text{Status}(\text{JLR} - \text{D4} - 1) &= \text{PublisherExecutableInterfaceUnavailable}, \\ \text{Status}(D4; T_{D4}^{\text{complete}}) &= \text{NotReady}_{\text{JLR} - \text{D4} - 1}. \end{aligned}$$

Therefore the strongest established D4 endpoint is

$$\text{FullBaseContinuum}_{D4}^{\text{pub95-summary}}$$

for the explicitly weaker summary target, together with a fail-closed `NotReady` theorem for the complete cosmological target. The result is neither a fresh arbitrary-history likelihood closure nor a unique carrier-selection theorem.

Formal, summary, and executable objects are non-identical

The source and physical theorems establish semantic D4 objects. The released numerical record fixes the scalar band B_{95}^{pub} . The summary target uses only that band and its exact provenance. A complete cosmological likelihood would require a total theory-to-observable-to-likelihood map and remains gated by `JLR - D4 - 1`. No implication among these levels is assumed merely because their numerical coordinates can be written in one chart.

Contents

1	Introduction and exact theorem posture	1
2	Frozen dependency interface and no-backflow lock	2
3	Target system, public record, summary target, and complete-target boundary	4
4	SOE-D4-1 source universe and candidate-complete raw lifts	6
5	Guarded universal standard-thermal compiler	7
6	Universal one-point physical realization and retained transparency	11
7	Released scalar marginal band and ordering-resolved pullback	23
8	Public-95 summary-target lift fibres	25
9	Parameter descent	28
10	No-repair source composition and complete-realization dependence	29
11	Exact public-95 summary-target base image and standing split	30
12	Overlap, descent, and global reconstruction	31
13	Total scientific disposition and failure provenance	33
14	Named fresh joint-likelihood gate: <code>JLR-D4-1</code>	35

15	Exact PCR-5E handoff	37
16	Capstone theorem package	37
17	Physical interpretation and cross-phenomenon comparison	39
18	Empirical and computational certificate record	40
19	Hostile-referee closure map	43
20	Reopening ledger	44
21	Conclusion	45
A	Guarded standard-thermal compiler normal form	46
B	Exact public scalar-band mass-sum certificate	47
C	Primary observational packet and covariance certificate	47
D	JLR-D4-1 validation specification	48
E	Status and failure-certificate schema	48
F	Machine-readable theorem and handoff records	49
G	Notation summary	49

1 Introduction and exact theorem posture

The PCR-5D sequence asks how additional physical phenomena act on the exact neutrino-oscillation standing-carrier space. The base is the target-relative SCIC quotient $Q_{\text{osc}}^{\text{SCIC}}$, not a list of preferred ultraviolet models. A new target may eliminate base classes, preserve them without new standing, split them by new standing, or leave a typed unresolved remainder. PCR-5D4 applies that calculus to cosmological population and transport.

The ordinary statement that cosmology constrains a neutrino mass sum is not yet a carrier theorem. A carrier theorem must specify the population law, background, collision or decay history, perturbation hierarchy, observable quotient, target interface, retained-base restriction, fibre geometry, and failure semantics. It must also distinguish a released statistical summary from a fresh likelihood evaluator.

The hostile sequence isolated two distinct source-theoretic questions that had previously been conflated: S4-04 proves an all-null conservative expansion, while the positive D4 branch requires a partially active seed; and source-fixed activity data are not automatically freely assignable activity data. Version 1.5.0 therefore inserts SAR-D4-1 between KSD-D4-1 and STR-D4-1. SAR-D4-1 proves, by direct finite inspection of the frozen SOE-D4-1 v1.0.0 source signature, that the standard-thermal seed-active/off-seed-null Boolean activity vector is itself a coherent source model: activity maps are source parameters and guards, no frozen constructor or constitutive clause generates one activity value from another, and the relative extension functor exists before kinetic saturation is evaluated. STR-D4-1 then applies KSD-D4-1 at that already coherent partial activity stratum, and D4-03 remains restricted to the representative family actually needed by the fibre theorem. No likelihood reconstruction, proof assistant, verified Boltzmann solver, or representative-independent canonical section is introduced.

1.1 The central structural result

The positive target is now an explicitly weaker finite summary object,

$$T_{D4}^{\text{pub95-summary}} = T_{D4}^{\text{pub95-summary}}.$$

Its empirical content is exactly the released scalar band B_{95}^{pub} . Its non-empirical content requires source formation, one-point physical realization, retained transparency, exact target identity, approximation control, and provenance. Relative to this target, every oscillation base class admits a continuum of normal-ordering lifts.

The complete cosmological target is separate:

$$T_{D4}^{\text{complete}} = T_{D4}^{\text{complete}}.$$

It requires the unavailable fresh correlated joint-DESI evaluator and therefore remains `NotReadyJLR-D4-1`. The summary theorem does not repair or complete that target.

The developing D-series pattern is therefore:

Extension	Base image	New standing effect
D1 direct kinematics	$Q_{\text{osc}}^{\text{SCIC}}$	absolute support scale
D3 successor active closure	proper nonempty subset of $Q_{\text{osc}}^{\text{SCIC}}$	active-process closure
D4 public-95 summary	$Q_{\text{osc}}^{\text{SCIC}}$	scalar mass-sum refinement inside a fixed thermal envelope

The D4 result is a universal split and fibre restriction, not a base discriminator.

1.2 Four levels of claim

The paper maintains four levels of authority.

1. *Semantic source level*. SOE-D4-1 forms the candidate history universe before any data response.
2. *Physical realization level*. A common semantic apex must satisfy all inherited, pure-D4, cross-incidence, common-cone, conservation, and continuation equations.
3. *Public-summary target*. The exact released scalar band defines $T_{D4}^{\text{pub95-summary}}$; this target carries no posterior-density or fresh-likelihood semantics.
4. *Complete target*. A total fresh theory-to-joint-likelihood map would define T_{D4}^{complete} ; this remains blocked by JLR–D4–1.

No statement is transported upward without a named theorem.

1.3 Proof architecture

Sections 2–4 freeze the dependency, target, source, and no-backflow interfaces. Sections 5–7 construct the guarded compiler, least kinetic closure, exact stress-energy descent dichotomy, seed-active relative extension, seeded transparent section, mass-indexed background family, one-point physical apex, and D4-only transparency ideal. Sections 8–9 define the scalar summary band, compute its ordering-resolved set-theoretic pullbacks, and activate the exact scalar standing output. Sections 10–14 determine the summary-target fibres, descent, declared-resolution source profile, base image, overlap, and failure boundary. Section 15 fixes JLR–D4–1; Section 16 exports the two-level D4 handoff; and Sections 17–21 give the capstone, interpretation, evidence boundary, hostile map, and reopening conditions.

2 Frozen dependency interface and no-backflow lock

2.1 Imported theorem package

The imported results and local burdens are summarized in Table 1. None of the imported results proves the D4 endpoint by itself.

Table 1: Frozen dependency interface and local D4 burden.

Source	Imported result	New burden in this paper
PCR-3	Candidate-complete source-first carrier universe, typed incidence, activity, and source-lawful maps.	Construct and evaluate the D4 population-and-transport source extension without modifying the old source object.
PCR-4	One-point simultaneous physical constraint collapse and total scientific status semantics.	Construct a single population-background-perturbation apex and retain exact failure provenance.
PCR-5B	Exact mixed-moduli oscillation standing-carrier atlas $\mathcal{N}(T_{\text{osc}})$ and quotient $Q_{\text{osc}}^{\text{SCIC}}$.	Determine the D4 restriction image and connected lift fibres without relabelling family charts as base classes.

Source	Imported result	New burden in this paper
PCR-5C	Source reduct, standing restriction, no merger, connected homotopy fibres, splitting, source sufficiency, and handoff calculus.	Prove every D4-specific hypothesis: target completeness, physicality, transparency, response totality, fibre disposition, and overlap.
SOE-D4-1	Candidate-complete finite-presentation, function-valued population-and-transport source universe with exact old-language reduct.	Construct the universal standard-thermal compiler and identify the participating target branches.
PCR-5D1	Full base image and continuum splitting by direct absolute-support standing.	Determine whether D4 merely restricts that scale fibre or creates a new base selector.
PCR-5D3	Proper base image at successor active-closure target 1.3.	Compare the final D4 image and separator without inferring a common generator.
Transport-closure paper	Seven-coordinate type (A, W, Q, C, B, E, M) , formal/executable separation, hidden-remainder audit.	Descend each D4 point to the certificate interface without using the certificate to generate source histories.
Payload completion	Exact primary CMB initialization; verified DESI archive, covariances, and posterior ensemble; publisher executable gap.	State and enforce the public/executable branch boundary and the JLR gate.

The frozen dependency order is documented in the upstream source, physical-collapse, standing-atlas, target-extension, and phenomenon-extension manuscripts [1, 2, 3, 4, 6, 7]. The seven-coordinate inference discipline is imported from the companion transport-closure analysis [8]; the local proof below uses that object only after source and physical realization.

2.2 Frozen base object

Let $\mathcal{N}(T_{\text{osc}})$ be the standing-core groupoid of the fixed oscillation target and

$$Q_{\text{osc}}^{\text{SCIC}} = \pi_0 \mathcal{N}(T_{\text{osc}}).$$

For $q \in Q_{\text{osc}}^{\text{SCIC}}$, let $\mathcal{N}(T_{\text{osc}})[q]$ denote the full connected subgroupoid representing q . Every D4 object carries a B15-framed source reduct and the complete retained oscillation packet. Distinct base classes retain all PCR-5B separators.

Theorem 2.1 (No-backflow lock). *No D4 source constructor, target acceptance rule, summary-target status, fibre disposition, or standing separator may depend on the later D4 endpoint, on the D1 full-image theorem, or on the D3 proper-image theorem. D1 and D3 may enter only after the D4 image and separator have been independently computed.*

Proof. SOE-D4-1 forms the source universe before target response. PCR-5C requires target identity and restriction to be fixed before fibre evaluation. If a D4 constructor or acceptance rule depended on a desired D1-like or D3-like endpoint, the candidate domain or

target would change after inspection of its image, violating source-first independence and the no-characterization-backflow theorem. The only lawful use of D1 and D3 is therefore as a post-calculation comparison tag. $\square$

2.3 Standing carrier identity

Two D4 presentations represent the same enriched standing carrier only when their complete D4 standing cores agree after lawful source, target, nuisance, approximation, and presentation descent. Family name, ultraviolet familiarity, field count, numerical grid, and posterior-chain index are not standing classifiers. Conversely, an admitted difference in mass sum, ordering response, population history, free-streaming scale, decay history, or observable continuation remains standing-visible when it survives the complete target.

3 Target system, public record, summary target, and complete-target boundary

3.1 Target branches

A D4 branch is a tuple

$$\beta = (b_{\text{bg}}, b_{\text{hist}}, b_{\text{data}}, b_{\text{resp}}, b_{\text{like}}, b_{\text{app}}),$$

containing a background model, population history, data packet, response map, likelihood or summary rule, and approximation certificate. Stable thermal free streaming, variable radiation density, nonstandard population, interaction or recoupling, decay or injection, and background extensions are distinct branches.

The source and physical branch used below is

$$\beta_0 = (\Lambda\text{CDM, stable thermal free streaming, Planck + ACT + DESI}_{\text{public}}, \text{standard Boltzmann, } \mathfrak{A}_0).$$

It does not yet choose between a scalar summary and a complete likelihood.

3.2 Three non-identical target objects

Definition 3.1 (Public-release record). $T_{D4}^{\text{public release}}$ is the exact retained public record: released bytes, covariance objects, configurations, nuisance and calibration ledgers, posterior ensembles, and their documented interpretation. It is evidence, not by itself a pointwise acceptance relation on arbitrary new theory objects.

Let μ_{pub} be the released scalar mnu marginal and define

$$\Sigma_{95}^{\text{pub}} := q_{0.95}(\mu_{\text{pub}}) = 0.061934524 \text{ eV}, \quad B_{95}^{\text{pub}} := (-\infty, \Sigma_{95}^{\text{pub}}].$$

Definition 3.2 (Public-95 summary target). The summary target $T_{D4}^{\text{pub95-summary}}$ consists of:

1. the frozen base target and branch β_0 ;
2. a source-faithful one-point D4 physical realization;
3. in-situ retained-oscillation transparency;
4. exact target, approximation, and provenance records;

5. the scalar-valued summary evaluator

$$S_{\Sigma}(W) := \Sigma_{\nu}(W) \in \mathbb{R}_{\geq 0};$$

6. the Boolean acceptance rule

$$\text{Acc}_{95}(W) = \mathbf{1}_{B_{95}^{\text{pub}}}(S_{\Sigma}(W)).$$

The exact scalar value is retained as a target-authorized standing output; the Boolean rule is its threshold projection. This is not an additional empirical conjunct: the scalar evaluator is the already required input of the frozen membership predicate. The target contains no posterior-density, relative-likelihood, joint-credible-region, evidence, or ordering-conditioned probability semantics.

Definition 3.3 (Complete cosmological target). T_{D4}^{complete} additionally requires a validated total map from every admitted fresh theory point to the correlated target observables and joint likelihood, including covariance, nuisance, prior, mask, approximation, and failure semantics.

Theorem 3.4 (Summary-target identity, independence, and non-equivalence). *The identity of $T_{D4}^{\text{pub95-summary}}$ is determined by β_0 , the independently retained scalar record μ_{pub} , the fixed functional $q_{0.95}$, the scalar evaluator S_{Σ} , and the source/physical/transparency predicates. It is independent of $Q_{\text{osc}}^{\text{SCIC}}$, the D4 fibre calculation, and the desired endpoint. At the current release,*

$$T_{D4}^{\text{pub95-summary}} \neq T_{D4}^{\text{complete}}.$$

In particular, summary compatibility does not imply a fresh likelihood value or complete-target survival.

Proof. The data defining μ_{pub} , its quantile functional, the branch record, and the scalar map S_{Σ} were fixed before the carrier image is evaluated. The scalar map is required to form the predicate $\mathbf{1}_{B_{95}^{\text{pub}}} \circ S_{\Sigma}$; making its exact output explicit changes no acceptance condition. None of these data contains a base-class label, source-family tag, or endpoint predicate. The target rule is therefore determined without reference to whether its image is empty, proper, or full. The complete target contains a total evaluator absent from the summary target and from the public release; hence no current target equivalence exists. $\square$

Theorem 3.5 (D4-01: target-interface completeness). *For the frozen D4 programme:*

1. *the Planck low- ℓ TT and EE, NPIPE CamSpec TTTEEE, and ACT DR6 plus Planck lensing packet is byte-complete and executable;*
2. *the public joint DESI packet is complete at its released archive, covariance, and posterior-record interface;*
3. *$T_{D4}^{\text{pub95-summary}}$ is a fully specified finite summary target with endpoint-independent identity;*
4. *the complete fresh joint-DESI evaluator is absent, so*

$$\text{Status}(\text{JLR} - \text{D4} - 1) = \text{PublisherExecutable}^{\text{InterfaceUnavailable}}.$$

$$\text{Status}(D4; T_{D4}^{\text{complete}}) = \text{NotReady}_{\text{JLR-D4-1}}.$$

5. *any change of background, history, data packet, response, summary functional, likelihood, nuisance, covariance, or approximation outside a proved target equivalence defines a successor target.*

Proof. The payload-completion audit fixes 1,782 primary-CMB files and verifies the selected CMB likelihoods. The joint DESI archive, separately released covariance arrays, and selected scalar posterior record have exact byte identities and initialization records. The release contains no complete theory-to-ShapeFit/BAO-to-log-likelihood implementation. The summary target is nevertheless total because its empirical predicate is the explicitly frozen scalar band. The complete target remains undefined on fresh points until JLR–D4–1 closes. $\square$

Status 3.6 (Current target statuses).

$$\begin{aligned} \text{Status}(T_{D4}^{\text{public release}}) &= \text{PublicReleaseComplete} \\ &\quad \text{InitializedAtPublishedInterface}, \\ \text{Status}(T_{D4}^{\text{pub95-summary}}) &= \text{IdentityFrozen} \\ &\quad \text{SummaryRuleComplete}, \\ \text{Status}(T_{D4}^{\text{complete}}) &= \text{NotReady}_{\text{JLR-D4-1}}. \end{aligned}$$

3.3 Seven-coordinate inference certificate

Every physically realized target-facing D4 point descends to

$$\tau_{\text{cos}} = (A, W, Q, C, B, E, M),$$

where A fixes the background anchor, W contains population and transport witnesses, Q is the admitted observable or summary quotient, C is continuation, B is the boundary and calibration ledger, E is the summary or likelihood rule, and M is the hidden-remainder coordinate. The projection is downstream of source and physical formation. The seven-coordinate discipline separates the formal theorem object, the finite released summary, and a future executable representative; none is used to manufacture another.

4 SOE-D4-1 source universe and candidate-complete raw lifts

4.1 Source signature

SOE-D4-1 extends the frozen PCR-3/B15 language by the sorts

PopulationState, PhaseSpaceDistribution, ThermalHistory, CollisionKernel,
DecayInjectionKernel, MetricBackgroundReference, PerturbationHierarchy,
CosmologicalObservableResponse, InitialDataCertificate, ApproximationCertificate.

Its operations include population, thermalization, partial population, free streaming, collision, decay, injection, redshift transport, perturbation, and projection to CMB, BAO, full-shape, and lensing observables.

For a B15-framed base source H , write

$$P_{H,\eta} \in \mathfrak{U}_{\nu,D4}^{\text{src}}$$

for a participating D4 history and

$$U_{D4}(P_{H,\eta}) = H$$

for its exact old-language reduct.

4.2 Candidate completeness

Theorem 4.1 (D4-02: candidate-complete raw lift domain). *Let $q \in Q_{\text{osc}}^{\text{SCIC}}$ and choose any B15-framed source presentation H of q . Every participating finite-presentation, function-valued D4 semantic history over H has a representative in $\mathfrak{U}_{\nu, D4}^{\text{src}}$, and every source term interprets as a declared semantic history. The quotient by lawful source redescription is equivalent to the semantic history groupoid. Every generated point is therefore present in the raw D4 lift domain before target evaluation.*

Proof. This is the SOE-D4-1 source-universe equivalence applied fibrewise over the exact reduct H . Semantic trace extensionality records every standing-relevant source object and relation. The typed source signature generates the complete trace, the normal-form theorem removes constructor bracketing and enumeration artifacts, and the lawful quotient identifies exactly the semantic equivalences. The exact reduct preserves every old node, source equation, primitive and higher-arity incidence, parameter namespace, automorphism, applicability record, and failure field. Hence no lawful D4 history is omitted from the raw domain and no generated source term is accepted without a semantic interpretation. $\square$

4.3 Activity reduction

D4 activity reduction is applied after source formation:

$$P_{H,\eta}^{\text{act}} = \text{ActRed}_{D4}(P_{H,\eta}).$$

It removes declared source-null D4 constructors but preserves all active incidences and constitutive equations. Fully source-null expansions witness conservativity and do not become duplicate D4 lifts.

Corollary 4.2 (No source-induced base merger). *If two D4 source histories have nonisomorphic PCR-3/B15 reducts in $\mathcal{N}(T_{\text{osc}})$, then they cannot become one enriched D4 standing carrier while the frozen base packet remains defined.*

Proof. The exact reduct is functorial and retains every base separator. An enriched isomorphism would restrict to an isomorphism of the base standing cores, contradicting their membership in different connected components. $\square$

5 Guarded universal standard-thermal compiler

5.1 Oscillation-compatible mass shapes

Fix the rounded oscillation projection values

$$\delta := \Delta m_{21}^2 = 7.5 \times 10^{-5} \text{ eV}^2, \quad \Delta := |\Delta m_{3\ell}^2| = 2.5 \times 10^{-3} \text{ eV}^2.$$

They are used only to define mass-sum maps for the summary target. For lightest mass $m_0 \geq 0$, define

$$\Sigma_{\text{NO}}(m_0) = m_0 + \sqrt{m_0^2 + \delta} + \sqrt{m_0^2 + \Delta}, \tag{1}$$

$$\Sigma_{\text{IO}}(m_0) = m_0 + \sqrt{m_0^2 + \Delta - \delta} + \sqrt{m_0^2 + \Delta}. \tag{2}$$

These maps preserve the frozen oscillation splittings and introduce no ultraviolet-family selector.

Lemma 5.1 (Strict monotonicity). *Both Σ_{NO} and Σ_{IO} are continuous and strictly increasing on $[0, \infty)$.*

Proof. Each square-root term is continuous. For $a > 0$,

$$\frac{d}{dm_0} \sqrt{m_0^2 + a} = \frac{m_0}{\sqrt{m_0^2 + a}} \geq 0.$$

Each mass-sum derivative contains the strictly positive derivative of the explicit m_0 term and two nonnegative terms. $\square$

5.2 Retained propagation seed and complete kinetic closure

For a B15-framed base source H , let $\mathbf{P}_{\text{osc}}(H)$ denote its complete retained propagation interface: the target-visible active modes, fixed splitting and mixing packet, boundary instruments, continuation, matching, and source map already evaluated by PCR-4/PCR-5B. Its object support is the seed

$$\mathbf{S}_0(W_H) := \text{supp } \mathbf{P}_{\text{osc}}(H).$$

Fix a coherent SOE-D4-1 source history over the framed physical source W_H . A *D4 kinetic forcing relation* is a source-active typed relation in its complete activity-reduced trace whose declared semantic effect is one of:

1. production, thermalization, partial population, free streaming, or collision;
2. decay, invisible loss, daughter production, or injection;
3. a hidden or environmental attachment carrying population, kinetic, or stress-energy work;
4. a constituent, binding, derivation, profile, or collective relation carrying an explicit D4 population or stress-energy map;
5. a source-to-stress-energy map.

Let

$$\text{Cl}_{\text{src}}^{D4}(W_H)$$

be the least constitutive closure of the complete activity-reduced source trace under the frozen PCR-3/SOE-D4-1 derivation rules. Thus a D4 kinetic consequence is included whether it is primitive or source-theoretically derived; no target evaluator or posterior verdict enters this closure. Write $\mathcal{R}_{\text{kin}}^{D4}(W_H)$ for the finite family of D4 kinetic forcing relations in $\text{Cl}_{\text{src}}^{D4}(W_H)$, and write $\partial^- r, \partial^+ r$ for their typed required-input and output supports.

Fix a coherent D4 source-parameter point η . For every $r \in \mathcal{R}_{\text{kin}}^{D4}(W_H)$, define the history-dependent source enabling predicate

$$\text{En}_r(S; W_H, \eta) = 1$$

iff every non-null required input of r lies in S , the relation-specific source activity-support map satisfies $a_r(\eta) = 1$, and the frozen coupling, threshold, memory, orientation, and initial-data equations of r are satisfied at η . Source-autonomous relations have empty required-input support. The activity map a_r is source-fixed data of the declared history family, not an output inferred from a later target response. This predicate therefore evaluates the actual source point: a nonzero old coupling or an open kinematic threshold is retained, but it generates a D4 populated output only when the corresponding typed D4 relation itself is active and its complete enabling equations hold.

A purely old PCR-3 relation with no primitive or derived D4 kinetic codomain remains fully present in the old source packet, but it is not silently converted into a D4 production or stress-energy map. If a claimed physical implication requires a conversion not generated by $\text{Cl}_{\text{src}}^{D4}(W_H)$, the correct status is **NotReady** or **SourceUpdate**, not a null D4 verdict.

Define inductively

$$\begin{aligned} \mathbf{S}_{n+1}(W_H; \eta) &:= \mathbf{S}_n(W_H; \eta) \cup \bigcup \left\{ \partial^+ r : r \in \mathcal{R}_{\text{kin}}^{D4}(W_H), \right. \\ &\quad \left. \text{En}_r(\mathbf{S}_n(W_H; \eta); W_H, \eta) = 1 \right\}, \\ \mathbf{S}_0(W_H; \eta) &:= \mathbf{S}_0(W_H). \end{aligned}$$

Because the source presentation and its frozen constitutive closure are finite, this sequence stabilizes. Its least fixed point is

$$\text{KinSat}_{D4}(W_H; \eta) := \mathbf{S}_N(W_H; \eta) \quad (N \leq |\text{Ob}(H)| + |\text{Ob}_{D4}|).$$

When the coherent history point is fixed, the manuscript abbreviates this by $\text{KinSat}_{D4}(W_H)$. The D4 participation predicate of that history is then

$$a_H^{D4} = 1 \text{ on } \text{KinSat}_{D4}(W_H; \eta), \quad a_H^{D4} = 0 \text{ on its D4-participation complement.}$$

The second clause concerns only newly adjoined D4 participation. It never deactivates, deletes, or relabels an old PCR-3 source node or source-active relation.

Lemma 5.2 (Typed non-generation, derived-consequence saturation, and naturality). *The old PCR-3/B15 language has no codomain in a D4 population, collision, decay/injection, perturbation, response, or stress-energy sort. Consequently an old source relation enters $\mathcal{R}_{\text{kin}}^{D4}(W_H)$ only when the frozen source theory supplies an explicit primitive or derived SOE-D4-1 incidence with that codomain. In particular, if an old coupling together with the frozen constitutive rules entails a D4 production or stress-energy consequence, that consequence lies in $\text{Cl}_{\text{src}}^{D4}(W_H)$ and is evaluated by En_r . Moreover, a lawful B15/D4 source isomorphism $\psi : (W_H, \eta) \rightarrow (W_{H'}, \eta')$ preserving the activity-support maps induces*

$$\psi_* \text{KinSat}_{D4}(W_H; \eta) = \text{KinSat}_{D4}(W_{H'}; \eta').$$

Proof. The typed-extension theorem S4-03 adjoins the D4 sorts and requires every relation between an old attachment and a new D4 object to be explicit and typed. Closing under the frozen derivation rules captures every source-theoretic D4 consequence of those typed relations. The exact reduct S4-09 preserves the complete old trace and forgets only the explicit new symbols. Thus no old-language arrow can acquire a D4 codomain by target interpretation alone, while no typed derived consequence is omitted.

The conservative null expansion of S4-04 is a model of the extended source theory over every old source model. Therefore the old source theory alone cannot entail a nonzero off-support D4 kernel or stress contribution: such an entailment would have to hold in the null expansion and would contradict its defining equations. A lawful isomorphism preserves the constitutive closure, activity guards, coupling and threshold equations, relation typing, endpoints, enabling predicates, and the retained propagation seed. Induction on n gives $\psi_* \mathbf{S}_n(W_H; \eta) = \mathbf{S}_n(W_{H'}; \eta')$, hence equality at the finite fixed point. $\square$

Remark 5.3 (Exact scope of source faithfulness). Source faithfulness is evaluated against the complete frozen trace χ_{D4} and its constitutive closure, not against an undeclared rule that converts every old coupling into a cosmological collision operator. Every old source-active coupling, threshold, ownership, memory, constituent, collective, and higher-incidence datum remains in the trace and must receive one of three exact D4 dispositions: propagation/stress-energy descent, an exactly closed support, or an explicit independent kinetic landing. A claim that an old relation necessarily generates a missing production,

collision, decay/injection, or stress map is a falsifier of the current source theory and has status SourceUpdate; it is never silently treated as D4-null. No numerical small-coupling or large-threshold approximation deletes an active source relation.

Lemma 5.4 (Guarded null-participation legitimacy). *Let η be a coherent D4 history point and let x be an old source object or incidence outside $\text{KinSat}_{D4}(W_H; \eta)$. Assigning the declared null or terminal D4 object to x and setting $a_H^{D4}(x) = 0$:*

1. *satisfies every new D4 constitutive relation incident on x ;*
2. *leaves every old PCR-3/B15 equation and active relation unchanged;*
3. *introduces no equation forcing a new old-language consequence;*
4. *preserves all primitive and higher-arity old incidence.*

Proof. By fixed-point closure at η , no enabled primitive or derived D4 kinetic forcing relation has all of its required inputs in $\text{KinSat}_{D4}(W_H; \eta)$ and an output at x . SOE-D4-1 S4-03 requires every remaining relation that could constrain an old attachment through a new D4 object to be guarded by the D4 activity predicate and to admit the declared null or terminal object. S4-04 proves conservativity by this activity-guarded expansion-and-reduct argument. Hence the new D4 relations are satisfied where their complete enabling predicate is false, while the entire old source presentation remains literally present. The lemma nulls only the new D4 participation leg, not the old source law, coupling, threshold, memory, or higher incidence. $\square$

5.3 Compiled thermal source object

The compiled standard-thermal history

$$\text{CTH}_\omega(H, m_0)$$

is formed as follows:

1. retain H , its full active relation packet, and $\text{P}_{\text{osc}}(H)$ without deletion;
2. seed three effective population slots on the three target-visible propagation modes, with masses determined by $\omega \in \{\text{NO}, \text{IO}\}$ and m_0 , preserving the frozen splittings and mixing coordinates;
3. retain and type every explicit source-active D4 kinetic or stress-energy incidence and compute $\text{KinSat}_{D4}(W_H)$ from the complete trace;
4. assign the standard Fermi-Dirac population and weak-decoupling history to the seeded slots;
5. assign the guarded null D4 object only on the complement of $\text{KinSat}_{D4}(W_H)$, while retaining all old source activity and relations;
6. form the full source-authorized D4 stress-energy diagram, including every populated or explicitly stress-bearing object in $\text{KinSat}_{D4}(W_H)$;
7. attach the flat- Λ CDM background reference, transport, perturbation hierarchy, response interfaces, and approximation certificate;
8. include the target-projection, population-to-stress-energy, hierarchy-to-response, and provenance diagrams;
9. evaluate the transparent/nontransparent dichotomy before physical-apex and target evaluation.

The compiler does not identify the effective population slots with the primitive occupants of every ultraviolet realization. It attaches the thermal envelope to the exact target-visible propagation interface and then allows source-active D4 relations, if present, to enlarge the kinetic closure.

Theorem 5.5 (Standard-thermal compiler theorem). *For every B15-framed base source H , $\omega \in \{\text{NO}, \text{IO}\}$, and $m_0 \geq 0$, $\text{CTH}_\omega(H, m_0)$ is a well-typed SOE-D4-1 source history satisfying*

$$U_{D4}(\text{CTH}_\omega(H, m_0)) = H.$$

It preserves every old source equation, primitive incidence, higher-arity incidence, and base automorphism; it carries the complete source-active D4 kinetic trace into $\text{KinSat}_{D4}(W_H)$; and it is natural under lawful B15 source isomorphism. The theorem does not yet assert that $\text{KinSat}_{D4}(W_H) = \mathbf{S}_0(W_H)$; that is the burden of KS–D4–1.

Proof. All constructors and incidences are in the authorized SOE-D4-1 signature. Lemma 5.2 supplies the complete typed forcing trace and its natural least closure. Lemma 5.4 licenses the null extension only outside that closure. The exact reduct deletes only new D4 symbols and returns the full old source presentation. A lawful B15 isomorphism transports the retained seed, every old incidence, every explicit D4 kinetic relation, and therefore the fixed-point closure; the generic thermal construction yields an isomorphic D4 source history. $\square$

Proposition 5.6 (Outcome independence). *Changing B_{95}^{pub} , deleting the released response record, replacing the background target, changing JLR–D4–1, or changing the eventual D4 image does not change $\text{CTH}_\omega(H, m_0)$ as a source construction.*

Proof. The compiler uses only the frozen base source, the ordering and mass parameters, and the source-authorized thermal/background constructors. Summary and likelihood outcomes occur only in downstream target status. $\square$

6 Universal one-point physical realization and retained transparency

6.1 Complete residual partition

Let W_H be a framed PCR-4/B15 physical realization of H . For a compiled lift, partition the complete D4 residual family as

$$\mathcal{E}_{D4} = \mathcal{E}_{\text{inh}} \sqcup \mathcal{E}_{\text{pop}} \sqcup \mathcal{E}_{\text{bg}} \sqcup \mathcal{E}_{\text{cross}} \sqcup \mathcal{E}_{\text{apex}}.$$

The five blocks contain, respectively: inherited PCR-4/B15 equations; population, kinetic, collision-null, and conservation equations; background, stress-energy, and perturbation equations; source-population-background-response commuting relations; and one-point common-cone, continuation, and matching equations.

The standard thermal envelope over $\mathbf{P}_{\text{osc}}(W_H)$ is the usual homogeneous Fermi-Dirac background and collisionless Einstein-Boltzmann hierarchy with the masses $M_\omega(m_0)$. Its physical fields are denoted $W_{\text{th}}(m_0, \omega)$. The combined object is constructed explicitly by retaining every carrier and relation of W_H , adjoining the fields of W_{th} , and identifying only their common target-visible propagation slots through the compiler attachment maps. No marginal witness from another source point is imported.

6.2 Mass-indexed flat- Λ CDM background closure

The phrase “fixed flat- Λ CDM branch” fixes the model class and anchor ledger; it does not assert that one numerical background solution is reused for every mass. Let

$$\mathfrak{A}_0^{\text{bg}} = (H_0, \omega_b, \omega_c^0, \omega_\gamma, T_{\gamma 0}, T_{\nu 0}, \Omega_\Lambda^0, \kappa_{\text{bg}})$$

be the frozen background anchor. The retained primary-CMB initialization witness supplies

$$\omega_c^0 = 0.12121379$$

at the normal-ordering floor.

The hierarchy certificate κ_{bg} distinguishes two non-identical realization modes:

$$\text{mode}(\kappa_{\text{bg}}) \in \{\text{SemanticHierarchy}, \text{CertifiedFiniteTruncation}\}.$$

The endpoint proved here fixes

$$\text{mode}(\kappa_{\text{bg}}) = \text{CertifiedFiniteTruncation}.$$

Its certificate specifies ℓ_{max} , the momentum quadrature, the closure map, an error envelope ε_{tr} , and a validation ledger $\mathcal{V}_{\text{tr}}$; it licenses only the response statements covered by those fields. The function-valued phase-space distribution and collisionless Liouville transport remain semantic objects. An untruncated Einstein–Boltzmann response belongs to the alternative semantic mode only when κ_{bg} additionally supplies a weighted Banach space $\mathcal{X}_{\text{EB}}$, a common dense domain, an adiabatic initial datum, and a time-dependent generator satisfying the hypotheses for a unique strongly continuous evolution family. No finite truncation is identified with that semantic hierarchy, and an untruncated response without this well-posedness certificate is `NotReady`. Let $\rho_{\text{crit},100}$ denote the critical density associated with $100 \text{ km s}^{-1} \text{ Mpc}^{-1}$, let $n_{\nu 0}^{(1)}$ be the present number density of one standard thermal mass eigenstate including its antiparticle, and define

$$\zeta_\nu := \frac{\rho_{\text{crit},100}}{n_{\nu 0}^{(1)}} = 93.14 \text{ eV}$$

in the frozen standard-thermal approximation certificate [11, 12].

For ordering ω and lightest mass m_0 , let

$$\omega_{\nu 0}^\omega(m_0) = \frac{\rho_{\nu 0}^\omega(m_0)}{\rho_{\text{crit},100}}$$

be the present physical neutrino density obtained from the standard Fermi–Dirac phase-space integral, and define the compensating cold-dark-matter coordinate

$$\omega_c^\omega(m_0) = \omega_c^0 - \left[\omega_{\nu 0}^\omega(m_0) - \omega_{\nu 0}^{\text{NO}}(0) \right].$$

Thus the present total $\omega_c + \omega_\nu$ anchor is preserved while the neutrino mass varies. Define the lawful background domain

$$\mathfrak{D}_{\text{bg}} = \{(\omega, m_0) : m_0 \geq 0, \omega_c^\omega(m_0) > 0\}.$$

For use in the background theorem, write

$$I_{\text{NO}}^{\text{pub95}} := \left\{ m_0 \geq 0 : \Sigma_{\text{NO}}(m_0) \leq \Sigma_{95}^{\text{pub}} \right\}.$$

Section 7 proves that this set is exactly $[0, m_\star]$.

Lemma 6.1 (Cold-dark-matter margin on the positive summary branch). *The entire normal-ordering summary interval lies in $\mathfrak{D}_{\text{bg}}$. More precisely,*

$$0 \leq \omega_{\nu 0}^{\text{NO}}(m_0) - \omega_{\nu 0}^{\text{NO}}(0) \leq \frac{\Sigma_{95}^{\text{pub}} - \Sigma_{\text{NO}}(0)}{\zeta_{\nu}} < 3.52 \times 10^{-5}$$

for every $m_0 \in I_{\text{NO}}^{\text{pub}95}$, and hence

$$\omega_c^{\text{NO}}(m_0) > 0.1211786.$$

Proof. For fixed comoving momentum p , the map $m \mapsto \sqrt{p^2 + m^2}$ has derivative in $[0, 1]$. Therefore the increase of the present energy density of one thermal species is bounded by its number density times its mass increase. Summing the three species gives

$$0 \leq \rho_{\nu 0}^{\text{NO}}(m_0) - \rho_{\nu 0}^{\text{NO}}(0) \leq n_{\nu 0}^{(1)} [\Sigma_{\text{NO}}(m_0) - \Sigma_{\text{NO}}(0)].$$

Division by $\rho_{\text{crit},100}$ yields the displayed ζ_{ν} bound. On the summary interval, $\Sigma_{\text{NO}}(m_0) \leq \Sigma_{95}^{\text{pub}}$. Substitution of the frozen scalar endpoint and floor gives the numerical margin. The margin is over three orders of magnitude smaller than ω_c^0 , so the compensated cold-dark-matter density stays strictly positive. $\square$

For $(\omega, m_0) \in \mathfrak{D}_{\text{bg}}$, define

$$\rho_{\nu}^{\omega}(a; m_0) = \frac{gT_{\nu 0}^4}{2\pi^2 a^4} \sum_{i=1}^3 \int_0^{\infty} \frac{q^2 \sqrt{q^2 + a^2 m_{i,\omega}(m_0)^2 / T_{\nu 0}^2}}{e^q + 1} dq$$

and define $p_{\nu}^{\omega}(a; m_0)$ by the corresponding standard pressure integral. Let

$$H_{\omega}^2(a; m_0) = H_0^2 \left[\Omega_{\gamma} a^{-4} + \Omega_b a^{-3} + \Omega_c^{\omega}(m_0) a^{-3} + \frac{\rho_{\nu}^{\omega}(a; m_0)}{\rho_{\text{crit},0}} + \Omega_{\Lambda}^0 \right].$$

Because the compensated present matter sum is fixed, the same Ω_{Λ}^0 enforces flatness at $a = 1$.

Theorem 6.2 (BG-D4-1: Mass-Indexed Flat- Λ CDM Background Closure). *For every $(\omega, m_0) \in \mathfrak{D}_{\text{bg}}$, the preceding data determine a mass-indexed background-and-perturbation object*

$$\mathcal{B}_{\omega, m_0} = (a_{\omega, m_0}, H_{\omega}(\cdot; m_0), \rho_{\nu}^{\omega}, p_{\nu}^{\omega}, \omega_c^{\omega}, \Delta_{\omega, m_0}, \kappa_{\text{bg}})$$

such that:

1. the flat Friedmann equation, standard component conservation equations, and collisionless neutrino Liouville equation hold;
2. if $\text{mode}(\kappa_{\text{bg}}) = \text{SemanticHierarchy}$, the adiabatic linear Einstein-Boltzmann problem has a unique mild solution in $C([\tau_0, \tau_1], \mathcal{X}_{\text{EB}})$ on every compact target interval; if $\text{mode}(\kappa_{\text{bg}}) = \text{CertifiedFiniteTruncation}$, the certified finite hierarchy has a unique ODE solution and every exported response retains $(\varepsilon_{\text{tr}}, \mathcal{V}_{\text{tr}})$;
3. $(\omega, m_0) \mapsto \mathcal{B}_{\omega, m_0}$ is continuous on compact subsets of $\mathfrak{D}_{\text{bg}}$ in the selected semantic norm or finite-coordinate norm;
4. H_0 , the present total matter anchor, the radiation ledger, flatness, and the target model class remain fixed, while $\omega_c^{\omega}(m_0)$ is the deterministic compensating continuation coordinate;

5. *the construction is natural under lawful B15 base-source isomorphism because it depends only on the retained propagation packet, the mass record, and $\mathfrak{A}_0^{\text{bg}}$.*

In particular, every $m_0 \in I_{\text{NO}}^{\text{pub95}}$ has a lawful flat- Λ CDM background in the same frozen model branch.

Proof. The phase-space integrals are finite and continuous in $a > 0$ and the mass parameters by dominated convergence. Collisionless transport gives

$$\dot{\rho}_\nu + 3H(\rho_\nu + p_\nu) = 0.$$

Lemma 6.1 gives strict positivity of the compensated cold-dark-matter density on the full positive summary interval; the definition of $\mathfrak{D}_{\text{bg}}$ gives it on the general domain. The right-hand side of the expanding-branch equation

$$\dot{a} = aH_\omega(a; m_0)$$

is positive, continuous, and locally Lipschitz for $a > 0$, so the background initial-value problem has a unique solution on each compact target interval. In semantic-hierarchy mode, the generator and common-domain hypotheses recorded in κ_{bg} give a unique strongly continuous evolution family and hence a unique mild solution. In certified-finite mode, the frozen truncation and closure rule give a finite linear ODE system with a unique solution; only certificate-covered responses are exported. Continuous parameter dependence follows from the corresponding linear-evolution theorem in semantic mode, from the standard ODE dependence theorem in finite mode, and from dominated convergence of the neutrino moments. The compensation formula preserves the present total matter anchor and leaves the branch, flatness rule, and radiation ledger unchanged. No ultraviolet presentation enters these equations beyond the common retained propagation packet, which proves naturality. $\square$

6.3 Source-faithful kinetic saturation and stress-energy descent

Let $\mathcal{A}_H^{\text{act}}$ be the finite set of old source-active relations of W_H . A *D4 landing ledger* is a total assignment

$$\begin{aligned} \lambda_H : \mathcal{A}_H^{\text{act}} &\longrightarrow \mathbb{L}_{D4}, \\ \mathbb{L}_{D4} &:= \{\text{PropagationDescent}, \text{SupportClosed}, \\ &\quad \text{IndependentKinetic}, \text{UnresolvedLanding}\}. \end{aligned}$$

The entries mean, respectively: the relation factors through the retained matching/projection and has no independent D4 occupation; its D4 support is exactly empty by a source support equation; it lands in an independent populated, collisional, decay/injection, environmental, or stress-bearing D4 object; or the frozen source theory lacks the law required to decide the landing. SupportClosed is exact and may not be inferred from a small nonzero coupling or a merely large finite threshold. The ledger belongs to the complete trace, is preserved by lawful source arrows, and is evaluated before a component can be called D4-null.

Fix a background $\mathcal{B}$. Let $\mathbf{SE}_{D4}(\mathcal{B})$ be the additive category of finite typed stress-energy presentations over $\mathcal{B}$. An object is a triple

$$(V, \mathbf{t}, \tau),$$

where V is the finite free real module generated by a source-authorized family of stress-bearing contributions, $\mathbf{t} \in V$ is its total-contribution vector, and

$$\tau : V \longrightarrow \Gamma(S^2 T^* \mathcal{M})$$

is the source-certified symmetric stress-energy map. Arrows are source-incidence-induced linear maps preserving $\mathbf{t}$, τ , support, and orientation. This category has finite colimits, computed as finite direct sums modulo the relations generated by its arrows. Thus the construction below is a typed finite coequalizer, not an untyped appeal to “adding tensors.”

For a complete D4 source trace over W_H , let

$$\mathcal{T}_{D4}(W_H) : \mathcal{J}_H \longrightarrow \mathbf{SE}_{D4}(\mathcal{B})$$

be the finite diagram of every source-active D4 population-bearing or explicitly stress-bearing object. Constituent, binding, collective, projection, derivation, and hybrid-gluing arrows are retained in $\mathcal{J}_H$. Define its full contribution object and evaluated tensor by

$$\mathbf{E}_H^{\text{full}} := \text{colim}_{\mathbf{SE}_{D4}(\mathcal{B})} \mathcal{T}_{D4}(W_H), \quad T_{\mu\nu}^{H,\text{full}} := \tau_H^{\text{colim}}(\mathbf{t}_H^{\text{colim}}).$$

Multiple presentations of one physical contribution are therefore coequalized before the total vector is evaluated. The compiled three-mode contribution object $\mathbf{E}_H^{\text{compiled}}$ has total tensor

$$T_{\mu\nu}^{\text{compiled}} := \sum_{i=1}^3 T_{\mu\nu}[f_i, m_i].$$

A *stress-energy descent certificate* is a natural arrow

$$d_H : \mathbf{E}_H^{\text{full}} \longrightarrow \mathbf{E}_H^{\text{compiled}} \quad \text{in } \mathbf{SE}_{D4}(\mathcal{B}),$$

which is the identity on the three seeded propagation slots, coequalizes every constituent/collective, derivation, projection, and gluing presentation, and records every nonzero extra contribution rather than discarding it. An isomorphic descent implies the tensor identity

$$T_{\mu\nu}^{H,\text{full}} = T_{\mu\nu}^{\text{compiled}}.$$

Define the priority-ordered nontransparent disposition map

$$\text{Disp}_{D4}^{\text{nt}} \in \left\{ \begin{array}{l} \text{SourceUpdate, NotReady,} \\ \text{SameTargetFail, TargetUpdate,} \\ \text{ResponseOutOfScope, PublicMarginalBandOutOfBand} \end{array} \right\}.$$

as follows: a required law outside SOE-D4-1 gives SourceUpdate; a missing typed kernel, initial datum, stress map, or descent certificate gives NotReady; an inconsistent complete same-target physical residual system gives SameTargetFail; a lawful history requiring a changed background or response target gives TargetUpdate; a lawful physical history not evaluated by the released response gives ResponseOutOfScope; and a fully formed, response-applicable point whose exact scalar output lies outside the summary band gives PublicMarginalBandOutOfBand. The first applicable clause is the unique disposition.

Theorem 6.3 (KSD-D4-1: Source-Faithful Kinetic-Saturation and Stress-Energy Descent). *For every B15-framed physical source W_H and every coherent compiled D4 source history parameter η over it:*

1. *the least kinetic closure $\text{KinSat}_{D4}(W_H; \eta)$ exists, is finite, contains $S_0(W_H)$, and is natural under lawful B15/D4 source isomorphism;*
2. *every old source-active relation has exactly one landing-ledger value $\lambda_H(r)$; an unresolved landing is NotReady or SourceUpdate, never a null assignment;*
3. *exactly one of the following kinetic branches occurs.*

Transparent standard branch. *The source equations prove*

$$\text{KinSat}_{D4}(W_H; \eta) = S_0(W_H),$$

all off-support kernels vanish,

$$\mathcal{K}_{\text{prod}}^{\text{off}} = \mathcal{K}_{\text{coll}}^{\text{off}} = \mathcal{K}_{\text{dec}}^{\text{off}} = \mathcal{K}_{\text{inj}}^{\text{off}} = 0,$$

and the descent certificate is an isomorphism in the typed stress-energy category,

$$d_H : \mathbf{E}_H^{\text{full}} \xrightarrow{\cong} \mathbf{E}_H^{\text{compiled}}, \quad T_{\mu\nu}^{H,\text{full}} = T_{\mu\nu}^{\text{compiled}} = \sum_{i=1}^3 T_{\mu\nu}[f_i, m_i].$$

No source-active contribution is omitted and no constituent, collective, projected, or hybrid contribution is counted twice. Only in this branch does the standard-thermal compiler supply the source-faithful physical input used by D4-03.

Nontransparent source branch. *A source-active D4 relation strictly enlarges $\text{KinSat}_{D4}(W_H; \eta)$, an off-support kernel is nonzero, or the complete stress-energy diagram does not descend to the three compiled modes. The history is not called D4-null. It receives exactly $\text{Disp}_{D4}^{\text{nt}}$ and remains outside the transparent standard branch unless a later source/target theorem changes that disposition.*

Proof. Finite stabilization and naturality follow from Lemma 5.2. Trace extensionality and the source-first constitutive closure make λ_H total: every old active relation either has a typed D4 landing, an exact support-closure certificate, or a recorded unresolved landing. At the fixed point, every primitive or derived source-active D4 kinetic consequence whose complete enabling predicate is satisfied and whose required inputs are reachable from the target-visible seed has been included. Hence the equality $\text{KinSat}_{D4}(W_H; \eta) = S_0(W_H)$ is equivalent to the absence, at that same coherent source point η , of an enabled IndependentKinetic edge crossing from the seed closure to its complement. Lemma 5.4 then proves the four off-support zero equations as source equations rather than target nonapplicability statements.

The finite stress-energy diagram is formed in $\mathbf{SE}_{D4}(\mathcal{B})$ from every object that the complete constitutive closure declares populated or stress bearing. Its colimit is the explicit finite direct-sum quotient by all source-certified constituent, derivation, projection, collective, and gluing relations. Therefore an isomorphic descent to the three seeded thermal contributions proves both completeness and no double counting. If closure equality, off-support zeros, or descent fails, the point is nontransparent. Formation, readiness, physical consistency, target identity, response applicability, and scalar-band status are checked in the stated order, so exactly one fail-closed disposition applies. None of these alternatives follows from a target verdict alone. $\square$

Theorem 6.4 (SAR-D4-1: Seed-Active Relative Extension). *Fix the frozen SOE-D4-1 v1.0.0 source theory and let H be any B15-framed PCR-3 source presentation. Define the standard-thermal D4 subtheory over H ,*

$$S_{D4}^{\text{th}}(H) \subseteq S_{D4}(H),$$

to contain the literal old source H , the three target-visible population slots, their ordering-resolved mass records, normalized Fermi-Dirac phase-space distributions, the weak-decoupling initial-data packet, stable post-decoupling free-streaming operations, the BG-D4-1 background attachment, the certified finite-truncation hierarchy, the required response/provenance objects, and the declared null or terminal object of every off-seed D4 sort.

Then there is a lawful relative extension functor

$$\iota_{\text{th}} : \text{Mod}(\mathbf{S}_3)_{\text{B15}} \times I_{\text{NO}}^{\text{pub95}} \longrightarrow \text{Mod}(\mathbf{S}_{D4})$$

satisfying

$$U_{D4} \circ \iota_{\text{th}} = \text{pr}_1, \quad \text{Act}_{D4}(\iota_{\text{th}}(H, m_0)) = \mathbf{S}_0(W_H).$$

Equivalently, for every chosen B15-framed representative H_q there is a source-parameter map

$$j_q^{\text{th}} : I_{\text{NO}}^{\text{pub95}} \longrightarrow \Theta_{D4}^{\text{coh}}(H_q)$$

with

$$a_r(j_q^{\text{th}}(m_0)) = \begin{cases} 1, & r \in R_{\text{seed}}, \\ 0, & r \notin R_{\text{seed}}, \end{cases}$$

and the chosen Boolean activity vector is a solution of the complete frozen activity/coherence subsystem.

The load-bearing source-theoretic property is exact: in SOE-D4-1 v1.0.0 no constructor has an activity-support map as its codomain, and no frozen constitutive clause has the form

$$a_{\text{seed}} = 1 \implies a_{\text{off}} = 1$$

or otherwise derives one D4 activity-support value from another. Activity-support maps occur only as source-parameter data, pointwise active/null status, lawful-isomorphism-preserved fields, and guards on the D4 relations they control. Hence off-seed activity may be set to its declared null value while the seed is active, provided the corresponding null/terminal objects are used.

If a later same-scope source theorem adds or derives a cross-activity clause that forces an off-seed D4 activity map true, this theorem is reopened as a SourceUpdate; the positive section may not be restored by the S4-04 all-null expansion.

Proof. The proof is a finite schema audit of the frozen SOE-D4-1 source specification [5]; no target response, numerical posterior, or executable formalization enters.

First, the source signature fixes the new D4 constructors as

Populate, Thermalize, PartiallyPopulate, FreeStream, Collide,
Decay, Inject, RedshiftTransport, Perturb, Project_{*}.

None has an activity-support sort as codomain. Definition 6.1 of SOE-D4-1 instead places the activity-support maps in the source-parameter object of a declared history family, and Definition 9.1 states that every D4 incidence or process map carries a *source-fixed* activity-support map and a declared source-null class. S4-06 normalizes a coherent history only after the pointwise activity stratum has been instantiated and treats that stratum as part of the normal-form data preserved by lawful source isomorphism. Thus activity support is selected at source level; it is not generated by a later response or by another D4 operation.

Second, inspect the eleven frozen constitutive classes of S4-03. Their activity-bearing content is exhausted as follows:

Frozen constitutive class	Activity-support occurrence in the frozen source theory
support/endpoints/attachments	active maps require typed endpoints; inactive maps use the declared null/terminal endpoint; no activity map is produced
distribution positivity/normalization	constrains the distribution carried by an active population object; the null population object satisfies the terminal law
initial-data/population compatibility	constrains an active populated slot; no clause concludes another activity value
Liouville/Boltzmann evolution	constrains the active kinetic object; an inactive kinetic relation is guarded and carries the declared null object
collision invariants/conservation	constrains an active collision kernel; the zero/null kernel is the inactive value
decay/injection balance	constrains active decay/injection data; the stable null branch uses zero kernels
stress-energy/background coupling	constrains the typed active contribution diagram and the BG-D4-1 attachment; it does not generate activity support
hierarchy recursion/closure	constrains the declared active hierarchy/truncation object
response projection/covariance	constrains the response attached to an already formed physical/source object; it does not determine source activity
pointwise source activity/null status	records each a_r and its null class pointwise; the frozen class contains no inter-activity implication
approximation/error records	constrains only the finite-representation certificate downstream of the semantic activity stratum

The last row relevant to source activity is therefore genuinely pointwise. More formally, the activity/coherence subsystem has no primitive relation whose consequent or codomain is an activity-support map, and the complete constructor list contains no operation capable of generating one. The only cross-use of activity in S4-03 is as the guard on a relation that could otherwise constrain an old attachment through a new D4 object. Hence a clause such as $a_{\text{seed}} \Rightarrow a_{\text{off}}$ is not part of the frozen source theory.

Third, define $\iota_{\text{th}}(H, m_0)$ explicitly. Retain the old presentation H literally. On $S_0(W_H)$, activate the three standard thermal population slots and assign the ordering-resolved masses, normalized Fermi-Dirac distributions, weak-decoupling initial data, stable free streaming, BG-D4-1 background, and certified finite hierarchy/response packet. On every off-seed D4 sort, assign the source-null or terminal object and set the corresponding activity-support value to false. Since the activity/coherence subsystem is pointwise and the remaining constitutive equations are exactly the guarded or terminal equations listed above, this tuple satisfies every frozen source equation. Therefore

$$\iota_{\text{th}}(H, m_0) \in \text{Mod}(\mathbf{S}_{D4})$$

and its source parameter lies in $\Theta_{D4}^{\text{coh}}(H)$.

The exact reduct is immediate because only new D4 symbols were adjoined:

$$U_{D4}(\iota_{\text{th}}(H, m_0)) = H.$$

Pointwise activity reduction retains precisely the active seed D4 subdiagram and the protected old source, so its D4 activity support is $S_0(W_H)$. A lawful B15 source isomorphism transports the old presentation, seed, mass record, null objects, activity values, and all typed D4 attachments; the construction is therefore functorial. Restricting ι_{th} to a chosen representative H_q gives j_q^{th} .

Finally, the theorem is relative to the frozen source signature. If an additional same-scope law is later proved whose source semantics derives an off-seed activity from the active seed, that law is absent from the v1.0.0 signature just audited. The correct disposition is SourceUpdate, which reopens this theorem and its downstream uses rather than changing the activity vector by target fiat. $\square$

Theorem 6.5 (STR-D4-1: Seeded Transparent-Section Realizability). *For each $q \in Q_{\text{osc}}^{\text{SCIC}}$, choose one B15-framed physical representative W_{H_q} in the nonempty framed provenance fibre supplied by B15. There exists a source-parameter section*

$$\eta_q^{\text{th}} : I_{\text{NO}}^{\text{pub95}} \longrightarrow \Theta_{D4}^{\text{coh}}(H_q), \quad m_0 \longmapsto \eta_{q, m_0}^{\text{th}},$$

whose non-mass activity coordinates are constant in m_0 and whose D4 activity is the actual compiled partial assignment, not the S4-04 all-null assignment:

$$a_{D4} = 1 \text{ on } S_0(W_{H_q}), \quad a_{D4} = 0 \text{ off } S_0(W_{H_q}).$$

More precisely, at every $m_0 \in I_{\text{NO}}^{\text{pub95}}$:

1. every D4 constitutive equation of the frozen SOE-D4-1 source theory is satisfied at η_{q,m_0}^{th} ;
2. for every primitive or source-theoretically derived kinetic forcing relation with an off-seed output,

$$\text{En}_r(S_0(W_{H_q}); W_{H_q}, \eta_{q,m_0}^{\text{th}}) = 0;$$

3. consequently,

$$\text{KinSat}_{D4}(W_{H_q}; \eta_{q,m_0}^{\text{th}}) = S_0(W_{H_q});$$

4. at that same partially active source point,

$$\mathcal{K}_{\text{prod}}^{\text{off}} = \mathcal{K}_{\text{coll}}^{\text{off}} = \mathcal{K}_{\text{dec}}^{\text{off}} = \mathcal{K}_{\text{inj}}^{\text{off}} = 0;$$

5. every landing-ledger entry is resolved: UnresolvedLanding does not occur on the chosen section;
6. the complete stress-energy diagram admits an isomorphic descent

$$d_{H_q} : \mathbf{E}_{H_q}^{\text{full}} \xrightarrow{\cong} \mathbf{E}_{H_q}^{\text{compiled}}, \quad T_{\mu\nu}^{H_q, \text{full}} = \sum_{i=1}^3 T_{\mu\nu}[f_i, m_i];$$

7. every old PCR-3/B15 source node, equation, source-active relation, primitive incidence, irreducible higher-arity incidence, and framed provenance datum is retained literally under the exact reduct;
8. if a faithful same-scope source equation forbids such a seeded point for some q , the result is a genuine nontransparent/base-selection or source-update result. S4-04 is not invoked to restore the positive branch.

No representative-independent canonical section is asserted. Any family of choices

$$\chi : q \mapsto (W_{H_q}, \eta_q^{\text{th}})$$

satisfying the theorem is sufficient for the downstream restriction and fibre arguments.

Proof. Choose the B15-framed representative W_{H_q} . Apply Theorem 6.4 and define

$$\eta_{q,m_0}^{\text{th}} := j_q^{\text{th}}(m_0).$$

The source-theoretic burden that was previously missing is therefore settled *before* kinetic saturation is evaluated: η_{q,m_0}^{th} is a coherent model of the frozen SOE-D4-1 source theory at the actual partially active stratum, every seed activity value is true, every off-seed D4 activity value is false, and every frozen activity/coherence clause is satisfied. No inference from the S4-04 all-null expansion is used.

At this coherent point, every off-seed D4 forcing relation has its own source-fixed activity factor equal to zero. Hence

$$\text{En}_r(S_0(W_{H_q}); W_{H_q}, \eta_{q,m_0}^{\text{th}}) = 0$$

for every primitive or source-theoretically derived forcing relation with off-seed output. KSD-D4-1 therefore yields

$$\text{KinSat}_{D4}(W_{H_q}; \eta_{q,m_0}^{\text{th}}) = S_0(W_{H_q})$$

and Lemma 5.4 gives the four exact off-support kernel zeros at this same partially active point.

The landing ledger is total on the section. Target-visible old source relations land by PropagationDescent; exactly inactive D4 relations land by SupportClosed; and Theorem 6.4 excludes an unresolved activity-coherence clause. If a required new D4 codomain were missing from the frozen theory, the point would instead receive SourceUpdate or NotReady under KSD-D4-1.

The six PCR-5B strata retain their actual old source data. LAW retains process order, locality/memory, matching, threshold and target attachment. LOCAL retains primitive nodes, vertices, representations and incidences. HID retains nonzero visible-hidden coupling, hidden support, physical thresholds, dilation/projection, memory, ownership and orientation. COMP retains constituents, binding, derivation and ownership. SOL retains profile, topology, collective coordinates and persistence. HYB retains the complete common-cone gluing and irreducible higher-arity incidence. Theorem 6.4 establishes that these old data coexist coherently with the seed-active/off-seed-null D4 activity vector; KSD-D4-1 then evaluates, rather than assumes, their D4 kinetic landing.

At this source point no independently occupied or independently stress-bearing D4 object lies off the retained propagation support. The complete stress-energy diagram therefore consists of the three seeded propagation contributions together with old constituent, collective, projection, derivation and hybrid presentations whose landing values are PropagationDescent or SupportClosed. The frozen PCR-5B matching/projection arrows induce a functor from this full diagram to the three-mode compiled diagram that is the identity on the seeded generators and coequalizes every duplicate old presentation. Conversely each compiled generator is represented by its seeded source object. The two maps are inverse after the typed finite coequalizer, so

$$d_{H_q} : E_{H_q}^{\text{full}} \xrightarrow{\cong} E_{H_q}^{\text{compiled}}$$

and the tensor equality follows without omitted or double-counted source-authorized D4 stress energy.

Thus each chosen representative admits the required coherent seeded transparent source section. If a future same-scope source theorem adds a cross-activity implication or otherwise makes an off-seed D4 process mandatory, Theorem 6.4 is reopened and the affected component becomes a genuine source-update or base-selection case. S4-04 cannot be used to restore the positive section. $\square$

Theorem 6.6 (D4-03: Section-Parametrized Transparent Standard-Thermal Lift Theorem). *Fix any choice family*

$$\chi : q \mapsto (W_{H_q}, \eta_q^{\text{th}})$$

supplied by Theorem 6.5. Then, for every

$$q \in Q_{\text{osc}}^{\text{SCIC}}, \quad m_0 \in I_{\text{NO}}^{\text{pub95}},$$

there exists one semantic physical object

$$W_{H_q, m_0, \text{NO}}^{D4}$$

co-instantiating the complete chosen base carrier, the source-faithful seeded standard-thermal history $\eta_{q, m_0}^{\text{th}}$, the mass-indexed flat- Λ CDM background $\mathcal{B}_{\text{NO}, m_0}$, exact descended stress energy, collisionless transport, the certified finite-truncation perturbation hierarchy, and the response interfaces. On that object:

1. every residual in $\mathcal{E}_{\text{inh}}$ vanishes;
2. every residual in $\mathcal{E}_{\text{pop}}$ and $\mathcal{E}_{\text{bg}}$ vanishes;
3. every residual in $\mathcal{E}_{\text{cross}}$ and $\mathcal{E}_{\text{apex}}$ vanishes;
- 4.

$$\text{KinSat}_{D4}(W_{H_q}; \eta_{q, m_0}^{\text{th}}) = \mathbf{S}_0(W_{H_q}),$$

all four off-support kernel families vanish, $\mathbf{E}_{H_q}^{\text{full}} \cong \mathbf{E}_{H_q}^{\text{compiled}}$, and

$$T_{\mu\nu}^{H_q, \text{full}} = \sum_{i=1}^3 T_{\mu\nu}[f_i, m_i];$$

5. no old source-active relation is deleted, made inactive, or reclassified;
6. all primitive and irreducible higher-arity old source incidence is preserved.

If the chosen seeded pair is transported by a lawful B15/D4 isomorphism, the transported pair yields an isomorphic physical apex. No representative-independent canonical section of the base quotient is asserted or required.

Proof. Theorem 6.5 supplies, for the chosen representative of q , a coherent partially active source point whose kinetic saturation is exactly the retained propagation seed, whose off-support kernels vanish, and whose full stress-energy diagram descends once to the three compiled modes. The inherited block vanishes because W_{H_q} is a PCR-4/B15 physical realization.

The standard thermal distribution and weak-decoupling history satisfy the population, conservation, and collisionless Liouville equations on the seeded source point. Theorem 6.2 supplies, for the same mass record $m_0 \in I_{\text{NO}}^{\text{pub95}}$, a lawful background $\mathcal{B}_{\text{NO}, m_0}$ in the certified finite-truncation mode fixed by the endpoint target, on which the Friedmann, stress-energy conservation, and certificate-covered perturbation equations hold. The background-to-hierarchy and hierarchy-to-response squares therefore commute on the same mass-indexed object. The combined object is one semantic tuple with one shared propagation interface, one source-faithful seeded history, one exact stress-energy descent, one mass-indexed background, and one set of cross-incidence maps. All five residual blocks vanish on this one apex.

Transporting the chosen seeded pair along a lawful isomorphism preserves the source activity-support maps, the KSD closure, the background family, and the retained interface, so the transported construction is isomorphic. The proof uses only existence of one seeded section over each base component; it makes no claim that every B15 presentation of the same base class carries the identical enriched section. $\square$

6.4 D4-only ideal and no-return typing

Let $\mathcal{I}_{D4}^{\text{th}}$ be the target-relative ideal generated by the D4 population, thermal-history, background-reference, stress-energy, perturbation, response, approximation, and D4-only incidence fields of the compiled envelope.

Lemma 6.7 (No D4-to-old-output constructor). *In the compiled summary branch, every cross constructor with a D4 input has D4 output. No constructor has a D4-only input and a retained oscillation-output sort. The only old-to-D4 cross maps are the read-only target-projection and attachment maps. Moreover, changing m_0 preserves every frozen Δm^2 , mixing coordinate, old boundary instrument, matching map, deletion field, and applicability field.*

Proof. This is the typed compiler signature. The D4 envelope is attached downstream of the retained propagation interface. Any new constructor feeding D4 stress-energy or background information back into an old oscillation-output sort would define a different target/source branch and require a new transparency theorem. The ordering-adapted mass maps change the common absolute scale while preserving the fixed squared-mass differences and mixing packet. $\square$

Theorem 6.8 (D4-04A: in-situ retained-oscillation transparency). *Every compiled apex of Theorem 6.6 lies in the SOE-D4-1 transparent locus. With all D4 fields physically present,*

$$\Phi_4^{\text{ret}}(W_{H_q, m_0, \text{NO}}^{D4}) = \Phi_4(W_{H_q}), \quad \Xi_{\text{osc}}^{\text{ret}}(W_{H_q, m_0, \text{NO}}^{D4}) = \Xi_{\text{osc}}(W_{H_q}).$$

More explicitly,

$$\begin{aligned} T^{\text{ret}} &= T, & R^{\text{ret}} &= R, & \Delta_{\text{frdel}}^{\text{ret}} &= \Delta_{\text{frdel}}, & C_{\text{act}}^{\text{ret}} &= C_{\text{act}}, \\ M_{\text{match}}^{\text{ret}} &= M_{\text{match}}, & B_{\text{bdy}}^{\text{ret}} &= B_{\text{bdy}}, & F_{\text{same}}^{\text{ret}} &= F_{\text{same}}. \end{aligned}$$

Proof. By Lemma 6.7, every D4-only generator lies in $\mathcal{I}_{D4}^{\text{th}}$, every retained evaluator annihilates that ideal, and no mixed constructor with old oscillation output exists. The target and boundary records are unchanged. The propagation response depends on the frozen splittings and mixing packet, which the mass maps preserve. The old deletion, activity, matching, and same-scope fields are literally retained inside the common apex. Therefore every retained evaluator factors through the quotient by $\mathcal{I}_{D4}^{\text{th}}$ and returns the old channel value before any D4 field is forgotten. This is in-situ factorization, not deletion-based transparency. $\square$

Corollary 6.9 (D4-04B: restriction and no merger). *There is a restriction functor*

$$R_{D4}^{\text{phys, th}} : \mathcal{N}_{D4}^{\text{phys, th}} \longrightarrow \mathcal{N}(T_{\text{osc}})$$

induced by exact source reduct and in-situ transparency. It descends to SCIC quotients, and no two distinct base classes merge.

Proof. Restrict each enriched object and arrow to its exact old standing core. Functoriality follows from compiler and transparency naturality. An enriched isomorphism connecting distinct base components would restrict to a forbidden base isomorphism. $\square$

6.5 Seven-coordinate descent

Theorem 6.10 (Certificate-interface descent). *Every compiled physical point admits a record*

$$\Gamma_{D4}(W_{H, m_0, \omega}^{D4}, \mathfrak{A}_0) = (A, W, Q, C, B, E, M)$$

that separates source-derived population/transport data, the frozen background anchor and its deterministic mass-indexed continuation, the scalar summary quotient, continuation, boundary and calibration data, the summary rule, and any hidden remainder. The record is downstream of Theorem 6.6. It cannot create a source history, prove physicality, repair transparency, or supply the missing complete likelihood.

Proof. Each coordinate is read from an already formed physical point and frozen target record. Lawful source and approximation arrows transport the record naturally. The separation follows the declared seven-coordinate type discipline; no coordinate has inverse generative authority over the source or physical object. $\square$

7 Released scalar marginal band and ordering-resolved pull-back

7.1 Exact released scalar record

The selected released scalar record ensemble contains four chains, 348,099 rows, total stored weight 428,587, and weight-based effective sample size 291,843.86. Its released scalar m_{ν} coordinate has

$$\begin{aligned}\text{mean}(m_{\nu}^{\text{rel}}) &= 0.023283530 \text{ eV}, \\ \text{sd}(m_{\nu}^{\text{rel}}) &= 0.019603382 \text{ eV}, \\ q_{0.95}(m_{\nu}^{\text{rel}}) &= 0.061934524 \text{ eV}.\end{aligned}$$

The four-chain classic $\hat{R}$ diagnostic for the scalar coordinate is 1.000416. These are audit properties of the released record.

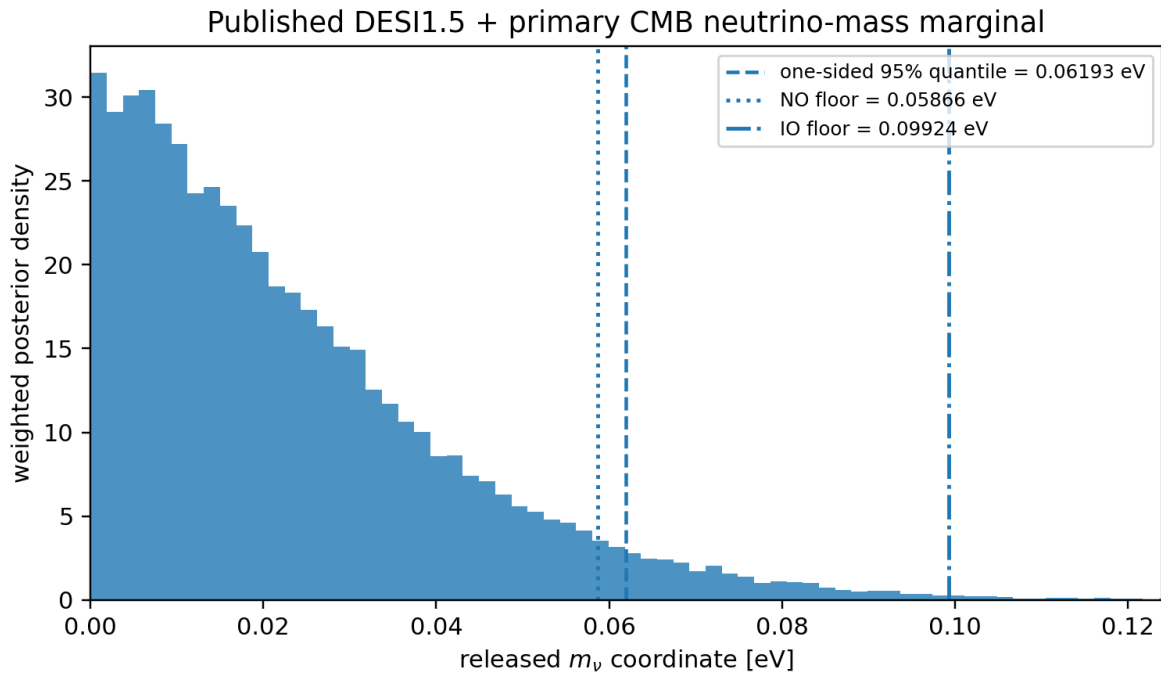


Figure 1: Weighted marginal of the released joint DESI plus primary-CMB scalar m_{ν} coordinate. The figure supplies the provenance of B_{95}^{pub} ; it is not an ordering-resolved posterior or a reconstructed likelihood.

Definition 7.1 (Released scalar marginal band).

$$B_{95}^{\text{pub}} := (-\infty, \Sigma_{95}^{\text{pub}}], \quad \Sigma_{95}^{\text{pub}} = 0.061934524 \text{ eV}.$$

A physically realized point of the summary branch receives `PublicMarginalBandCompatible` exactly when $\Sigma_{\nu} \in B_{95}^{\text{pub}}$, and `PublicMarginalBandOutOfBand` when $\Sigma_{\nu} \notin B_{95}^{\text{pub}}$. These are target-defined summary statuses, not statements about posterior density or fresh-likelihood satisfiability.

7.2 Mass-prescription separation

The released scalar coordinate contains samples below the physical three-mass normal-ordering floor. It therefore cannot be identified pointwise with either ordering-resolved map $\Sigma_{\omega}(m_0)$.

Theorem 7.2 (Mass-prescription and scalar-band transport theorem). *For each ordering ω , the only transport used in $T_{D4}^{\text{pub95-summary}}$ is the set map*

$$j_\omega : [0, \infty) \longrightarrow \mathbb{R}, \quad j_\omega(m_0) = \Sigma_\omega(m_0).$$

The summary-compatible parameter set is the set-theoretic pullback

$$I_\omega^{\text{pub95}} = j_\omega^{-1}(B_{95}^{\text{pub}}).$$

No posterior measure on m_0 , no ordering-conditioned credible interval, and no equality between the released scalar response and an exact three-mass cosmological response is asserted.

Proof. The empirical pass/fail rule of the summary target is defined only by scalar band membership, while the exact scalar value is retained separately as the standing output needed to form that rule. Applying the scalar map to a set requires no measure transport, likelihood equivalence, prior Jacobian, or fresh response. The observed sub-floor samples show why stronger identification would be invalid. Hence the theorem is exactly a pullback of sets and nothing more. $\square$

7.3 Ordering floors and exact pullbacks

Lemma 7.3 (Mass-sum floors).

$$\begin{aligned} \Sigma_{\text{NO}}(0) &= \sqrt{\delta} + \sqrt{\Delta} = 0.058660254 \text{ eV}, \\ \Sigma_{\text{IO}}(0) &= \sqrt{\Delta - \delta} + \sqrt{\Delta} = 0.099244289 \text{ eV}. \end{aligned}$$

Therefore

$$\Sigma_{\text{NO}}(0) < \Sigma_{95}^{\text{pub}} < \Sigma_{\text{IO}}(0).$$

Proof. Substitution in Equations (1)–(2) gives the values. The final inequality is direct comparison with the retained scalar band endpoint. $\square$

Theorem 7.4 (Normal-ordering scalar-band pullback). *There exists a unique $m_\star > 0$ defined exactly by*

$$\Sigma_{\text{NO}}(m_\star) = \Sigma_{95}^{\text{pub}}.$$

It obeys the certified enclosure

$$0.00276660960118537158 < \frac{m_\star}{\text{eV}} < 0.00276660960118537159,$$

and hence

$$m_\star \approx 0.002766609601185372 \text{ eV}.$$

Moreover,

$$I_{\text{NO}}^{\text{pub95}} = \Sigma_{\text{NO}}^{-1}(B_{95}^{\text{pub}}) = [0, m_\star].$$

This set has cardinality $\mathfrak{c}$.

Proof. Continuity, strict monotonicity, the lower-floor inequality, and divergence as $m_0 \rightarrow \infty$ give existence and uniqueness. Direct interval evaluation of the recorded radical expression gives

$$\Sigma_{\text{NO}}(0.00276660960118537158 \text{ eV}) < \Sigma_{95}^{\text{pub}} < \Sigma_{\text{NO}}(0.00276660960118537159 \text{ eV}),$$

which proves the enclosure without identifying the root with a terminating decimal. Monotonicity gives the exact preimage, and a nondegenerate real interval has cardinality $\mathfrak{c}$. $\square$

Theorem 7.5 (Inverted-ordering scalar-band disposition).

$$I_{\text{IO}}^{\text{pub95}} = \Sigma_{\text{IO}}^{-1}(B_{95}^{\text{pub}}) = \emptyset.$$

Every compiled standard inverted-ordering point therefore receives PublicMarginalBandOutOfBand under $T_{D4}^{\text{pub95-summary}}$.

Proof. Strict monotonicity and Lemma 7.3 give $\Sigma_{\text{IO}}(m_0) \geq \Sigma_{\text{IO}}(0) > \Sigma_{95}^{\text{pub}}$ for every $m_0 \geq 0$. $\square$

Exact boundary of the ordering statement

The theorem does not claim that inverted ordering is physically impossible, excluded by a fresh likelihood, or absent from the source universe. It says only that the standard stable-thermal inverted-ordering mass-sum image lies outside the explicitly frozen scalar summary band.

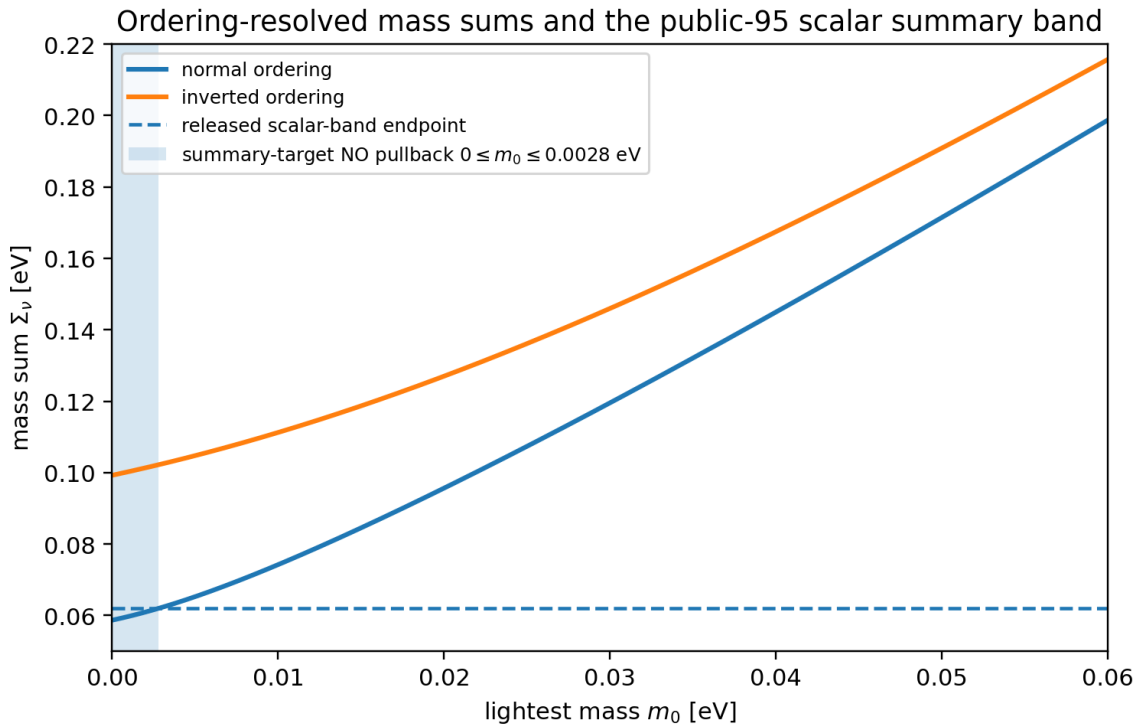


Figure 2: Ordering-resolved mass-sum maps and the frozen scalar summary band. The band intersects the normal branch at m_* and lies below the inverted-ordering floor. No posterior measure is transported by this diagram.

8 Public-95 summary-target lift fibres

8.1 Three target-indexed groupoids

The target distinction produces three different groupoids.

1. $\mathcal{N}_{D4}^{\text{phys,th}}$: source-faithful, one-point, retained-transparent standard-thermal physical realizations, independent of data status.

2. $\mathcal{N}_{D4}^{\text{pub95-summary}}$: those physical realizations satisfying the exact summary target $T_{D4}^{\text{pub95-summary}}$.
3. $\mathcal{N}_{D4}^{\text{complete}}$: realizations accepted by the complete cosmological target. This groupoid remains unavailable until JLR–D4–1 closes.

The corresponding restriction maps are not interchangeable.

Theorem 8.1 (SSA-D4-1: Summary-Scalar Standing Activation). *The summary target carries a scalar-valued evaluator*

$$S_{\Sigma} : \mathcal{N}_{D4}^{\text{phys,th}} \longrightarrow \mathbb{R}_{\geq 0}, \quad S_{\Sigma}(W) = \Sigma_{\nu}(W),$$

fixed before fibre evaluation. It has the following properties:

1. every lawful D4 standing isomorphism preserves S_{Σ} exactly;
2. the public-95 acceptance map factors as

$$\text{Acc}_{95} = \mathbf{1}_{B_{95}^{\text{pub}}} \circ S_{\Sigma};$$

3. the target identity retains the exact value of S_{Σ} , not only the Boolean value of Acc_{95} ;
4. therefore

$$S_{\Sigma}(W) \neq S_{\Sigma}(W') \implies W \not\cong W'$$

in the summary-target standing groupoid.

Proof. The exact mass record is already standing under PCR-5D1 and is retained by the D4 compiler. Definition 3.2 fixes S_{Σ} as the scalar output needed by the independently frozen threshold rule; no fibre result is used in that definition. A lawful source, physical, nuisance, or approximation arrow preserves the semantic mass record and hence its sum. The Boolean rule is literally the indicator of the fixed band composed with S_{Σ} . Since enriched standing identity is required to preserve every target-authorized standing output, an isomorphism can exist only between points with equal scalar values. $\square$

For $q \in Q_{\text{osc}}^{\text{SCIC}}$, define

$$\mathcal{L}_{D4}^{\text{pub95-summary}}(q) = \mathcal{N}_{D4}^{\text{pub95-summary}} \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q].$$

Its objects retain the complete summary-target standing core together with a base-core isomorphism; its arrows retain source, physical, summary-target, approximation, and provenance data.

8.2 Seeded summary-target section

Fix one choice family

$$\chi : q \longmapsto (W_{H_q}, \eta_q^{\text{th}})$$

from Theorem 6.5 and define

$$s_{\chi}(q, m_0) = [W_{H_q, m_0, \text{NO}}^{D4}] \quad (m_0 \in I_{\text{NO}}^{\text{pub95}}).$$

A different lawful seeded choice may produce a different enriched section, but every such section restricts to the same base component. No representative-independent canonical choice is required for existence, surjectivity, or the continuum lower bound.

Theorem 8.2 (D4-05: exact summary-target connected-fibre theorem). *For every $q \in Q_{\text{osc}}^{\text{SCIC}}$,*

$$I_{\text{NO}}^{\text{pub95}} \longrightarrow \pi_0 \mathcal{L}_{D4}^{\text{pub95-summary}}(q), \quad m_0 \longmapsto s_\chi(q, m_0)$$

is injective. Consequently,

$$|\pi_0 \mathcal{L}_{D4}^{\text{pub95-summary}}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}.$$

The compiled normal-ordering branch is an interval-indexed family of pairwise distinct connected components with inherited base isotropy. Additional lawful summary-target components remain coproduct refinements. Unevaluated histories remain outside the accepted summary groupoid with their exact ROOS, readiness, or update status.

Proof. Lemma 6.1 and Theorem 6.2 place the complete interval inside the lawful background domain. Theorems 6.5 and 6.6 therefore supply one physical point on the chosen seeded section for every parameter value, Theorem 6.8 places it over q , and Theorem 7.4 gives summary compatibility. If $m_0 \neq m'_0$, strict monotonicity gives

$$S_\Sigma(W_{H_q, m_0, \text{NO}}^{D4}) = \Sigma_{\text{NO}}(m_0) \neq \Sigma_{\text{NO}}(m'_0) = S_\Sigma(W_{H_q, m'_0, \text{NO}}^{D4}).$$

Theorem 8.1 then forbids a standing isomorphism between the two objects. Base automorphisms act within a fixed scalar fibre and do not identify unequal mass sums. $\square$

Corollary 8.3 (Universal summary-target split). *Every base class is split:*

$$|\pi_0 \mathcal{L}_{D4}^{\text{pub95-summary}}(q)| > 1 \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}.$$

8.3 Branchwise disposition matrix

Table 2: Exact branchwise D4 dispositions at manuscript version 1.5.0.

Branch	Target status	Fibre effect	Exact interpretation
Primary CMB	Byte complete; initialized	Complete response component	The selected Planck/ACT likelihoods initialize at one common point.
Flat Λ CDM, stable thermal, NO	Summary compatible for $m_0 \in I_{\text{NO}}^{\text{pub95}}$	Continuous nonempty component over every base	Positive branch used in D4-05 and D4-08.
Flat Λ CDM, stable thermal, IO	Summary out of band	No public-95 summary component	Not source deletion and not fresh-likelihood emptiness.
Complete correlated cosmological target	NotReady _{JLR-D4-1}	Complete fibres unavailable	Positive JLR status required.
Flat $w_0 w_a$ CDM, stable thermal	Target update / not ready	No transferred interval	Distinct background target.
Variable N_{eff}	Not ready	Exact fibre unavailable	Radiation-density variation is independently standing-bearing.
Partial or nonthermal population	ROOS or not ready	Typed unresolved component	Standard summary target is nonapplicable.
Self-interaction, recoupling, decay, injection, hidden radiation	ROOS or not ready	Typed unresolved component	Dedicated process and response branch required.
Astrophysical face	Not ready	No astrophysical fibre claim	A future complete packet defines a successor target.

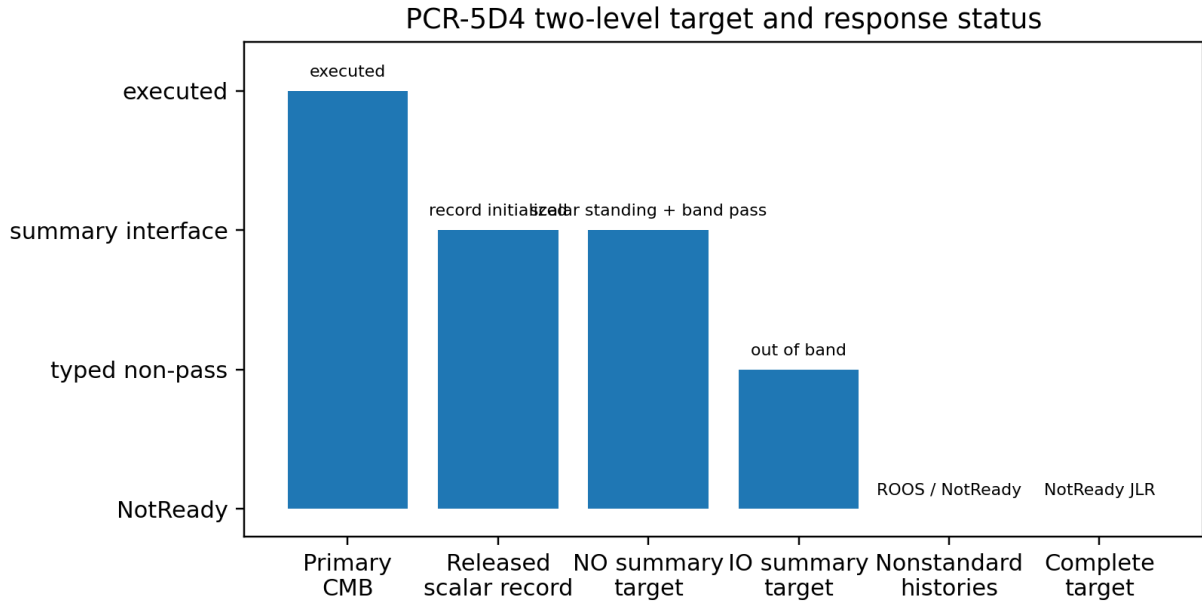


Figure 3: Two-level D4 status summary. The positive full-base theorem is attached to the public-95 summary target; the complete cosmological target remains JLR-gated.

9 Parameter descent

9.1 Parameter-role partition

For the compiled summary branch,

$$\Theta_{D4}^{\text{comp}} = \Theta_{\text{base}} \times \Theta_{\text{mass}} \times \Theta_{\text{pop}} \times \Theta_{\text{bg}} \times \Theta_{\text{summary}} \times \Theta_{\text{nuis}} \times \Theta_{\text{app}}.$$

Physical and standing roles are decided by the summary target and the one-point realization, not by numerical sampling alone.

Theorem 9.1 (D4-06: standing, nuisance, approximation, and presentation descent). *Two accepted summary-target compiled points represent the same D4 standing carrier exactly when:*

1. *their base restrictions are isomorphic in $\mathcal{N}(T_{\text{osc}})$;*
2. *their mass sum and ordering-resolved mass record agree;*
3. *their fixed thermal history, frozen background anchor, induced mass-indexed background continuation, and summary-band provenance agree;*
4. *their nuisance and calibration data are related by a target-authorized summary-preserving map;*
5. *their numerical representatives are connected by approximation certificates proving convergence to the same semantic history and target record;*
6. *their remaining difference is lawful source or presentation skin.*

A different grid, hierarchy truncation, chain index, or chart does not split a carrier, while unequal Σ_ν does.

Proof. Necessity follows from equality of complete summary-target standing cores. Sufficiency follows because the listed data exhaust the admitted summary target. Nuisance and approximation changes descend only through explicit target-preserving certificates; otherwise they are NotReady. $\square$

10 No-repair source composition and complete-realization dependence

10.1 Compiled macroblock resolution

For a compiled standard-thermal point x , define the three occupancy-bearing macroblocks

$$K_{\text{src}}^{(3)}(x) = \{\text{BaseCore}(x), \text{Population}(x), \text{KineticHistory}(x)\}.$$

The first entry is the complete already-classified PCR-5B base source, not a claim that every internal base block has one universal D4 deletion status. The second contains the relic population and mass occupancy; the third contains the thermal-history, free-streaming, and perturbation source laws. The mass-indexed background and scalar summary response are realization/target roles and are not counted as source macroblocks.

Let

$$N_{\text{src},x}^{(3)} : 2^{K_{\text{src}}^{(3)}(x)} \longrightarrow \{0, 1, \perp\}$$

be the fixed-parent, no-retuning sufficiency map at this declared resolution. Separately define

$$K_{\text{real}}^{(5)}(x) = K_{\text{src}}^{(3)}(x) \cup \{\text{BackgroundFamily}(x), \text{SummaryResponse}(x)\}$$

with no-retuning map $N_{\text{real},x}^{(5)}$.

For the fine PCR-5C source question, let $K_{\text{base}}^{\text{act}}(x)$ be the activity-reduced internal block set of the chosen base presentation and let

$$\text{ClSub}(x) \subseteq 2^{K_{\text{base}}^{\text{act}}(x) \sqcup K_{\text{D4}}^{\text{act}}(x)}$$

be its constitutively closed deletion domain. The present summary target does not evaluate that base-dependent fine hypergraph uniformly.

Theorem 10.1 (D4-07: compiled three-macroblock profile and realization-dependency theorem). *For every compiled standard-thermal summary-target survivor x ,*

$$N_{\text{src},x}^{(3)}(S) = 1 \iff S = K_{\text{src}}^{(3)}(x),$$

so the proved macroblock profile is

$$Y_{\text{compiled}, 3\text{-block}}.$$

Moreover,

$$N_{\text{real},x}^{(5)}(S) = 1 \iff S = K_{\text{real}}^{(5)}(x),$$

so the compiled complete realization has a uniquely minimal source/background/response dependency set at the five-role resolution. The exact fine-grained PCR-5C profile over $\text{ClSub}(x)$ is exported with status

$$\text{NotReady}_{\text{FineSourceDeletion}},$$

not silently identified with the three-macroblock result.

Proof. Removing BaseCore removes the retained carrier and its target-visible propagation interface; removing Population removes the occupied massive relic modes; removing KineticHistory removes the source-level thermal and free-streaming laws. At the declared macroblock resolution none can be restored by retuning, source replacement, regluing, or target change, so the full three-block set is uniquely minimal. At the complete-realization level, deleting BackgroundFamily removes the mass-indexed Friedmann and stress-energy equations, while deleting SummaryResponse removes target disposition.

These are genuine complete-realization dependencies but do not turn target-facing blocks into source constituents. Because internal PCR-5B active blocks and constitutive closure domains vary by stratum and presentation, no uniform fine-source profile follows from the macroblock deletion calculation. $\square$

Remark 10.2 (Intrinsic and fine-grained routes). An intrinsic source may carry additional population or response structure and may realize S, R, P, or Y when a branch-specific target evaluates it. Such routes are not overwritten by the universal compiler. The PCR-5E handoff therefore carries $Y_{\text{compiled,3-block}}$ together with the explicit fine-profile readiness status, rather than treating a macroblock profile as the exact PCR-5C source-sufficiency hypergraph.

11 Exact public-95 summary-target base image and standing split

Theorem 11.1 (D4-08: exact summary-target full-base-image theorem).

$$\text{im } \rho_{D4}^{\text{pub95-summary}} = Q_{\text{osc}}^{\text{SCIC}}.$$

Equivalently, the essential image of $R_{D4}^{\text{pub95-summary}}$ is all of $\mathcal{N}(T_{\text{osc}})$.

Proof. Fix $q \in Q_{\text{osc}}^{\text{SCIC}}$ and take the seeded pair $\chi(q) = (W_{H_q}, \eta_q^{\text{th}})$ supplied by Theorem 6.5. Theorem 7.4 contains $m_0 = 0$, and Lemma 6.1 places that point in the lawful background domain. Theorems 6.2 and 6.6 construct a one-point physical lift, Theorem 6.8 restricts it to q , and the summary rule gives PublicMarginalBandCompatible. Hence every base component lies in the essential image. The reverse inclusion is tautological. $\square$

Corollary 11.2 (Selection strength). *The summary target produces a universal fibre split and an ordering-sensitive scalar-band disposition, but no base-class elimination and no enriched singleton selection.*

11.1 Minimal separator and broader ledger class

Let the relative standing ledger include

$$\Sigma_\nu, \quad \omega, \quad f_i, \quad \mathcal{H}_{\text{bg}}, \quad k_{\text{fs}}, \quad \Pi_{\text{cos}}, \quad \tau_{\text{cos}}.$$

On the positive branch, the thermal history and background model class are fixed; the background solution varies only through the deterministic mass-indexed continuation of Theorem 6.2 and is not an independently free coordinate.

Theorem 11.3 (D4-09: split classes and exact separator theorem). *Every $q \in Q_{\text{osc}}^{\text{SCIC}}$ is split under $T_{D4}^{\text{pub95-summary}}$. For accepted normal-ordering points with $m_0 \neq m'_0$, the minimal proved primitive separator is*

$$S_{D4}^{\text{new,min}} = \Sigma_\nu, \quad \Sigma_{\text{NO}}(m_0) \neq \Sigma_{\text{NO}}(m'_0).$$

The broader response-ledger class is

$$\text{CosmologicalScaleHistory} = (\Sigma_\nu, \omega, \text{thermal/background/free-streaming context}).$$

The context is physically load-bearing for realization but is either fixed or deterministically induced by Σ_ν ; it is not claimed to be an independently minimal separator on this branch. The ordering result is a summary-target disposition through disjoint mass-sum floors, not an ordering-conditioned posterior theorem.

Proof. D4-05 gives the split. Strict monotonicity makes unequal mass sum sufficient and no smaller displayed primitive coordinate is needed. The remaining scale-history fields define the physical envelope and future variable-history comparison class, but they are fixed or deterministically induced rather than independently varied in the positive calculation; minimality cannot be inferred from them. $\square$

11.2 Comparison with D1 and D3

Corollary 11.4 (Comparison tags).

$$\text{Rel}_{D1}(D4) = \text{ScaleHistoryRefinement}, \quad \text{Rel}_{D3}(D4) = \text{DistinctSeparator}.$$

Proof. D1 made absolute support standing. D4 embeds that scale into a fixed cosmological population and transport envelope and restricts it through the scalar summary band, so it is a scale-history refinement even though Σ_ν is the minimal separator proved here. D3 uses an active-process closure condition and has a proper base image, whereas the D4 summary target has full base image. No common generator is inferred. $\square$

Exact two-level D4 endpoint

$$\begin{aligned} \text{Status}(D4; T_{D4}^{\text{pub95-summary}}) &= \text{FullBaseContinuum}_{D4}^{\text{pub95-summary}}, \\ \text{im } \rho_{D4}^{\text{pub95-summary}} &= Q_{\text{osc}}^{\text{SCIC}}, \\ |\pi_0 \mathcal{L}_{D4}^{\text{pub95-summary}}(q)| &\geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}. \end{aligned}$$

$$\text{Status}(D4; T_{D4}^{\text{complete}}) = \text{NotReady}_{\text{JLR-D4-1}}.$$

The first box is a theorem about the explicitly frozen public-95 summary target. The second box preserves the complete cosmological target as unresolved under JLR-D4-1.

12 Overlap, descent, and global reconstruction

12.1 Chart objects and semantic core maps

The compiled summary-target branch is covered by charts whose indices retain all data needed to decide standing identity:

$$i = (q, H, \omega, m_0, \eta_{\text{th}}, \mathcal{B}, \kappa, \alpha, \zeta_{\text{resp}}).$$

Here $q \in Q_{\text{osc}}^{\text{SCIC}}$, H is a B15-framed presentation of q , $\omega \in \{\text{NO}, \text{IO}\}$, m_0 is the lightest mass, η_{th} is the standard-thermal source packet, $\mathcal{B}_{\omega, m_0}$ is the mass-indexed background continuation, κ is an approximation certificate, α is a nuisance/calibration chart, and ζ_{resp} is a summary-target response certificate. Let Γ_i be the groupoid of points represented in chart i , with lawful source, parameter, nuisance, and approximation arrows. Each chart has a semantic standing-core map

$$s_i : \Gamma_i \longrightarrow \mathcal{N}_{D4}^{\text{pub95-summary, comp}}.$$

Definition 12.1 (Semantic presentation overlap). For two charts i, j , the presentation overlap is the homotopy fibre product

$$O_{ij} = \Gamma_i \times_{\mathcal{N}_{D4}^{\text{pub95-summary, comp}}}^h \Gamma_j.$$

For a finite nonempty index set $I = \{i_0, \dots, i_k\}$, the direct higher overlap is

$$O_I = \Gamma_{i_0} \times_{\mathcal{N}_{D4}^{\text{pub95-summary,comp}}}^h \dots \times_{\mathcal{N}_{D4}^{\text{pub95-summary,comp}}}^h \Gamma_{i_k}.$$

An object of O_I is a family of presentations together with coherent standing-core isomorphisms. Pairwise nonemptiness is not substituted for an object of O_I .

Definition 12.2 (Physical source amalgamation). A physical amalgamation is a typed source diagram

$$D_I \xrightarrow{\lambda_r} P_{H_r, \eta_r} \quad (r \in I)$$

in the authorized D4 source-map category, together with parameter-gluing, process-composition, common-background, and physical-apex certificates. It is not implied by a nonempty presentation overlap. The raw colimit is formed first and activity reduction is applied afterward:

$$P_I^{\text{raw}} = \text{colim}_{r \in I} (D_I \rightarrow P_{H_r, \eta_r}), \quad P_I = \text{ActRed}_{D4}(P_I^{\text{raw}}).$$

12.2 Chart-normalization invariance

Lemma 12.3 (Compiled chart normalization). *Every compiled public presentation admits a normalized chart representative obtained by:*

1. *canonical source-node and parameter namespacing;*
2. *deletion of source-null and identity D4 constructors at the instantiated activity point;*
3. *flattening of syntactic constructor nesting while retaining primitive higher-arity incidence;*
4. *canonical ordering of target, nuisance, and approximation certificate fields;*
5. *replacement of every lawful alpha-renaming by literal equality and every nontrivial invertible transport by a typed chart arrow.*

The normalized standing object is independent of enumeration order, constructor bracketing, momentum-grid naming, and chain-file index.

Proof. The source part is the SOE-D4-1 canonical-template theorem. D4-06 supplies nuisance and approximation descent. The base frame is unchanged by exact reduct, and the summary-target certificate records only target-bearing quantities and their lawful transports. Hence all listed presentation variations either normalize to equality or remain represented by an invertible arrow. $\square$

Theorem 12.4 (D4-10: source, history, branch, and approximation overlap closure). *For the compiled summary-target branch, the complete chart family $\{\Gamma_i\}_{i \in I_{D4}^{\text{pub,comp}}}$ satisfies:*

1. *every compiled public object lies in at least one chart;*
2. *an arrow of $\mathcal{N}_{D4}^{\text{pub95-summary,comp}}$ is represented by a chart arrow after passing to a pair overlap;*
3. *every lawful finite family of mutually compatible presentations is represented by its direct higher overlap O_I ;*
4. *two chart points represent the same standing carrier exactly under the D4-06 identity criterion;*
5. *physical source gluing occurs only through explicit typed amalgamation legs and commuting gluing certificates;*

6. the Cech nerve $\Gamma_\bullet$ of the cover has codescent

$$\text{Codesc}(\Gamma_\bullet) \simeq \mathcal{N}_{D4}^{\text{pub95-summary,comp}}.$$

Duplicate grids, chain files, B15 frames, and nuisance charts are therefore not counted as distinct carriers. The nonstandard-history remainder is retained separately as a typed ResponseOutOfScope/NotReady subfibration and is not silently absorbed into this codescent theorem.

Proof. Coverage follows from the standard-thermal compiler and D4-02. Let two chart points have the same semantic standing core. D4-06 gives a base isomorphism, equality of mass-sum, ordering, population, background, continuation, and observable records, together with authorized nuisance and approximation transports. These data define an object of O_{ij} and a chart arrow. Conversely, an overlap object carries a standing-core isomorphism and therefore satisfies the D4-06 identity criterion. Strict monotonicity of Σ_{NO} ensures that equal mass sum on the normal branch implies equal m_0 ; no hidden mass-coordinate multiplicity survives.

For higher overlaps the coherent isomorphisms are required directly. The raw physical source colimit is a separate construction from presentation descent and is formed only from typed source maps. Essential surjectivity of the chart cover and representation of every semantic arrow give the standard effective codescent equivalence for groupoids. None of these steps supplies a response for a history outside the compiled summary-target branch, so the unresolved remainder remains typed rather than deleted. $\square$

Corollary 12.5 (Grothendieck reconstruction of the compiled summary-target branch). *The restriction functor and its connected-base fibres reconstruct the compiled summary-target groupoid:*

$$\mathcal{N}_{D4}^{\text{pub95-summary,comp}} \simeq \int_{\mathcal{N}(T_{\text{osc}})} \mathcal{L}_{D4}^{\text{pub95-summary,comp}}.$$

At connected-component level,

$$Q_{D4}^{\text{pub95-summary,comp}} = \coprod_{q \in Q_{\text{osc}}^{\text{SCIC}}} \pi_0 \mathcal{L}_{D4}^{\text{pub95-summary,comp}}(q),$$

where the coproduct is disjoint because D4-04 preserves every base separator.

Proof. The restriction functor is no-merger by D4-04, and D4-05 supplies the connected fibre over every base component. Standard Grothendieck assembly gives the first equivalence. Passing to connected components yields the displayed disjoint union. $\square$

Remark 12.6 (Exact atlas boundary). The theorem is global for the compiled summary-target branch. It is not a global atlas for every function-valued SOE-D4-1 history. A nonthermal, interacting, decaying, injected, hidden-radiation, or enlarged-background history that lacks an authorized response remains in the semantic source universe but outside $\mathcal{N}_{D4}^{\text{pub95-summary,comp}}$. Its status is recorded by D4-11.

13 Total scientific disposition and failure provenance

13.1 Priority-ordered decision tree

Formation failure, target revision, response nonapplicability, scalar-summary disposition, and complete-target satisfiability belong to different stages.

Definition 13.1 (D4 status decision tree). For a proposed D4 point x , $\text{Status}_{D4}(x)$ is determined by the first applicable clause:

1. **SourceUpdate**: faithful construction requires a source sort, operation, relation, or law absent from SOE-D4-1;
2. **Undef**: the current language is used but the construction is malformed or ill typed;
3. **NotReady**: the source and target are fixed but a required map, packet, kinetic/stress-energy certificate, or complete interface is absent;
4. **SameTargetFail**: the source and target are fixed and well typed, but the one-point source/kinetic/stress-energy residual system fails before response evaluation; this is not a global fibre-emptiness theorem;
5. **TargetUpdate**: the source is lawful but the claimed target differs from the frozen target without an equivalence theorem;
6. **ResponseOutOfScope**: the point is lawful and physical but the frozen response is expressly nonapplicable to its history class;
7. **PublicMarginalBandCompatible**: the point satisfies the exact finite summary target $T_{D4}^{\text{pub95-summary}}$, including $\Sigma_\nu \in B_{95}^{\text{pub}}$;
8. **PublicMarginalBandOutOfBand**: the point is physically realized on the fixed branch but its scalar mass sum lies outside B_{95}^{pub} ;
9. **SolvedNonempty** or **SolvedEmpty**: only a complete, well-typed evaluator at fixed scope may assign semantic nonemptiness or emptiness.

No later clause is evaluated after an earlier clause applies.

Table 3: D4 status semantics and prohibited transfers.

Status	Exact meaning	Not licensed
Source update	New source language required.	Treating the history as physically absent.
Undefined	Malformed or ill-typed expression.	Evaluating satisfiability.
Target update	Lawful source under a different target.	Same-act repair.
Not ready	Required declared interface or certificate missing.	Physical exclusion or zero fibre.
Same-target fail	Complete one-point source/kinetic/stress residuals fail at the frozen target.	Global fibre emptiness, source deletion, or transfer to another target.
Response out of scope	Lawful history outside a response domain.	Applying the standard summary band.
Public marginal-band compatible	Exact summary-target pass.	Posterior density, evidence, fresh likelihood, or complete-target survival.
Public marginal-band out of band	Scalar sum outside the frozen band.	Physical impossibility, source deletion, or fresh-likelihood emptiness.
Solved empty/nonempty	Result of a complete evaluator at fixed scope.	Transfer to another target or history class.

Definition 13.2 (Failure certificate). A D4 unresolved or failure record is

$$\mathfrak{F}_x = (x, \beta, H, P_{H,\eta}, W_{H,\eta}^{D4}, \text{App}, \mathfrak{A}, \mathcal{R}, \text{Status}_{D4}(x), \text{Prov}).$$

A fibre may be declared empty only when every raw representative ends in **SolvedEmpty** under one complete evaluator.

Theorem 13.3 (D4-11: total failure, update, ROOS, summary, and readiness boundary). *The decision tree is mutually exclusive and exhaustive. Present consequences are:*

1. *every compiled normal-ordering point with $m_0 \in [0, m_\star]$ has status `PublicMarginalBandCompatible`;*
2. *every standard inverted-ordering compiled point in the lawful background domain has status `PublicMarginalBandOutOfBand`;*
3. *nonstandard histories lacking a validated branch response are `ResponseOutOfScope` or `NotReady` according to response applicability;*
4. *the complete target and all fresh joint-DESI fibres are `NotReady`_{JLR-D4-1};*
5. *the exact fine-grained PCR-5C source-deletion profile is `NotReady`_{FineSourceDeletion} and does not alter summary-target acceptance;*
6. *no ill-typed, not-ready, same-target-fail, ROOS, summary-out, fine-profile-unresolved, or JLR-blocked point is counted as `SolvedEmpty`.*

Every unresolved region carries a complete provenance record.

Proof. The clauses are ordered by construction stage. Source formation and typing precede background closure; background closure precedes physicality; target identity precedes readiness; readiness precedes applicability; summary status is evaluated only after physicality, transparency, and scalar standing activation; solved status is reserved for a complete evaluator. The consequences follow from BG-D4-1, KSD-D4-1, D4-03, D4-04, SSA-D4-1, the scalar-band pullbacks, the branch matrix, D4-01, and the resolution boundary of D4-07. $\square$

No inference from summary failure or unevaluability to exclusion

Neither

$$\Sigma_\nu \notin B_{95}^{\text{pub}}$$

nor absence of a standard-history response implies physical exclusion. The exact alternatives are summary out-of-band, `ResponseOutOfScope`, `NotReady`, `SourceUpdate`, or `TargetUpdate`, according to the first failed stage.

14 Named fresh joint-likelihood gate: JLR-D4-1

14.1 Exact missing executable object

The missing interface is not a data vector or covariance file. It is a total declared evaluator

$$\mathcal{L}_{D4,\text{joint}}^{\text{exec}} : \Theta_{D4}^\beta \longrightarrow \mathbb{R} \cup \{-\infty\}$$

that maps every admitted fresh theory point through the correlated DR1 full-shape and DR2 BAO observable construction, nuisance projection, covariance rule, prior, mask, approximation convention, and failure semantics. The public release contains a verified posterior ensemble and covariance evidence but does not expose this complete map.

14.2 Lawful completion routes

Definition 14.1 (JLR Route A). `PublisherInterfaceVerified` requires a publisher-supplied executable whose repository or release identity, code revision, data bytes, observable ordering, units, covariance convention, nuisance parameters, priors, masks, approximations, deterministic reference values, initialization, and released-posterior reproduction all pass the frozen validation ledger.

Definition 14.2 (JLR Route B). IndependentReconstructionCertified requires an independently constructed evaluator with:

1. target identity and no omitted observable block;
2. faithful theory-to-ShapeFit/BAO projection;
3. exact covariance ordering, normalization, and correlation treatment;
4. nuisance, prior, mask, and approximation equivalence;
5. reference-point likelihood agreement;
6. mock and covariance validation;
7. reproduction of the released posterior location and width within frozen tolerances;
8. stable derivatives and lawful parameter transport;
9. successful hostile-mutation tests fixed before D4 fibre evaluation.

Theorem 14.3 (JLR-D4-1: joint likelihood execution gate). *Fresh joint-DESI acceptance, fresh executable D4 fibres, and a fresh executable D4 base image are authorized if and only if*

$$\text{Status}(\text{JLR} - \text{D4} - 1) \in \{\text{PublisherInterfaceVerified}, \text{IndependentReconstructionCertified}\}.$$

At the present release,

$$\begin{aligned} \text{Status}(\text{JLR} - \text{D4} - 1) &= \text{PublisherExecutable,} \\ &\quad \text{InterfaceUnavailable,} \\ \text{Status}(\rho_{D4}^{\text{complete}}) &= \text{NotReady.} \end{aligned}$$

The public-95 summary theorems remain valid only for $T_{D4}^{\text{pub95-summary}}$; no complete-target fibre or base-image theorem is inferred from them.

Proof. A positive Route A or Route B status supplies the exact missing map together with target identity and validation, so fresh evaluation is defined. Conversely, a fresh D4 likelihood claim requires precisely such a defined and validated map; without one, the claim has no evaluator. The payload audit establishes the present absence. The summary theorem uses only the frozen scalar band and therefore neither assumes nor constructs the missing map. $\square$

Forbidden likelihood substitutions

None of the following closes JLR–D4–1: a posterior density estimator; interpolation over released samples; arbitrary-chain reweighting; a Gaussian fitted only to the published marginal; omission of off-diagonal covariance blocks; or a reconstruction tuned after inspecting the D4 survivor set. Each either changes the target or supplies no total theory-to-likelihood map.

14.3 Hostile validation requirements

Before Route B can become positive, the reconstruction must fail under each of the following mutations: permuted observable ordering, altered units, a removed covariance cross-block, undeclared covariance symmetrization, deleted nuisance coordinate, changed prior range, shifted mask, modified nonlinear prescription, broken theory projection, and posterior interpolation substituted for the likelihood. A reconstruction insensitive to any such load-bearing mutation is not faithful to the frozen target.

15 Exact PCR-5E handoff

15.1 Versioned two-level handoff object

Define

$$\begin{aligned} \mathcal{H}_{D4}^{1.5.0} = (&T_{D4}^{\text{public release}}, T_{D4}^{\text{pub95-summary}}, T_{D4}^{\text{complete}}, \mathfrak{U}_{\nu, D4}^{\text{src}}, \\ &\mathcal{N}_{D4}^{\text{pub95-summary}}, \mathcal{N}_{D4}^{\text{complete}}, R_{D4}^{\text{pub95-summary}}, \rho_{D4}^{\text{pub95-summary}}, \rho_{D4}^{\text{complete}}, \\ &\{\mathcal{L}_{D4}^{\text{pub95-summary}}(q)\}_{q \in Q_{\text{osc}}^{\text{SCIC}}}, \\ &I_{D4}^{\text{pub95, comp}}, S_{D4}^{\text{new, min}}, \\ &\text{ScaleHistoryLedger}_{D4}, \\ &\text{SrcProf}_{D4}, \text{RealizationDep}_{D4}, \\ &\text{Fail}_{D4}, \text{Overlap}_{D4}, \\ &\text{Approx}_{D4}, \tau_{\text{cos}}, \\ &\text{Rel}_{D1}, \text{Rel}_{D3}, \text{Status}(\text{JLR} - D4 - 1)). \end{aligned}$$

Theorem 15.1 (D4-12: exact PCR-5E handoff theorem). *The datum $\mathcal{H}_{D4}^{1.5.0}$ is sufficient to form two distinct PCR-5E marginal legs:*

1. *a positive public-95 summary leg with full base image and continuum fibres;*
2. *a complete cosmological leg with status $\text{NotReady}_{\text{JLR}-D4-1}$.*

The summary leg must retain its explicit target tag and may not be substituted for the complete leg. The handoff exports exact target identity, source universe, mass-indexed background closure, guarded physical lift, transparency, scalar-standing activation, fibres, minimal separator, broader ledger class, the declared-resolution source profile and fine-profile readiness status, realization dependency, overlap, failure provenance, approximation boundary, D1/D3 comparison tags, and JLR status. PCR-5E must still form one common source and physical pullback; marginal image intersection is not joint realization.

Proof. D4-01 separates the three target objects; D4-02 gives candidate completeness; BG-D4-1, KSD-D4-1, SAR-D4-1, STR-D4-1, and D4-03 give the mass-indexed source-faithful physical interface; D4-04 gives retained restriction; SSA-D4-1 and D4-05 give the standing-distinct summary fibres; D4-06 through D4-10 give descent, declared-resolution source profile, image, separator, and overlap; D4-11 gives total status; and JLR – D4 – 1 gives the complete-target boundary. These data determine two typed marginal legs without identifying them. $\square$

Marginal compatibility is not joint realization

Even though $\text{im}_{\rho_{D4}^{\text{pub95-summary}}} = Q_{\text{osc}}^{\text{SCIC}}$, a summary-target D4 lift over q need not share one source point, background, higher incidence, or physical apex with a D1, D2, D3, or D5 lift. PCR-5E must test those common-source and cross-phenomenon constraints directly.

16 Capstone theorem package

Theorem 16.1 (PCR-5D4 two-level cosmological population-and-transport closure). *Assume the frozen PCR-3/PCR-4/PCR-5B/PCR-5C package, SOE-D4-1 source authorization, the initialized public record, and the scalar-band certificate. Then:*

1. *D4-01 fixes $T_{D4}^{\text{public release}}$, $T_{D4}^{\text{pub95-summary}}$, and T_{D4}^{complete} as non-identical objects;*
2. *D4-02 includes every authorized participating source history before target evaluation;*
3. *BG-D4-1 constructs a lawful mass-indexed flat- Λ CDM background family over the full positive interval and distinguishes semantic-hierarchy from certified-finite-truncation authority;*
4. *KSD-D4-1 constructs the complete source-faithful kinetic saturation, proves the transparent/nontransparent dichotomy, and establishes exact no-omission/no-double-counting stress-energy descent;*
5. *SAR-D4-1 proves the partially active relative extension and solves the frozen activity/coherence subsystem at the compiled seed; STR-D4-1 constructs one transparent source section in every base component at that coherent point, and D4-03 constructs the corresponding one-point physical standard-thermal lift;*
6. *D4-04 proves in-situ retained transparency and no merger;*
7. *SSA-D4-1 activates the exact scalar output S_Σ before fibre evaluation;*
8. *D4-05 gives continuum summary-target fibres over every base class;*
9. *D4-06 gives target-correct nuisance, approximation, and presentation descent;*
10. *D4-07 proves $Y_{\text{compiled,3-block}}$, records exact five-role realization dependence, and keeps the fine PCR-5C profile readiness-gated;*
11. *D4-08 proves $\text{im } \rho_{D4}^{\text{pub95-summary}} = Q_{\text{osc}}^{\text{SCIC}}$;*
12. *D4-09 proves $S_{D4}^{\text{new,min}} = \Sigma_\nu$ and retains CosmologicalScaleHistory as the broader ledger class;*
13. *D4-10 reconstructs the compiled summary groupoid by direct overlap and codescent;*
14. *D4-11 assigns every raw history one exact formation, update, readiness, response, summary, or complete-evaluator status;*
15. *D4-12 exports the two distinct marginal legs to PCR-5E;*
16. *JLR–D4–1 remains negative, so no complete-target fibre or image claim is made.*

Consequently,

$$\boxed{\begin{aligned} \text{Status}(D4; T_{D4}^{\text{pub95-summary}}) &= \text{FullBaseContinuum}_{D4}^{\text{pub95-summary}}, \\ \text{im } \rho_{D4}^{\text{pub95-summary}} &= Q_{\text{osc}}^{\text{SCIC}}, \\ |\pi_0 \mathcal{L}_{D4}^{\text{pub95-summary}}(q)| &\geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}, \end{aligned}}$$

while

$$\boxed{\text{Status}(D4; T_{D4}^{\text{complete}}) = \text{NotReady}_{\text{JLR-D4-1}}.}$$

Proof. The numbered items are exactly Theorems 3.5-15.1, BG-D4-1, KSD-D4-1, the supporting scalar-transport theorems, and Theorem 14.3. The endpoint boxes add no empirical premise and preserve the target tags on which the conclusions depend. $\square$

Corollary 16.2 (No unique summary-target D4 realization). $T_{D4}^{\text{pub95-summary}}$ does not license a unique enriched carrier realization. Every base component has at least continuum many standing-distinct summary-compatible lifts, and all base components remain in the image.

Table 4: Exact two-level D4 endpoint matrix.

Scope	Final status	Mathematical meaning
Public-95 summary target	Full-base continuum	Full base image; continuum normal-ordering fibre over every base class.
Standard IO summary branch	Marginal-band out of band	Empty scalar-band pullback; no physical or complete-likelihood exclusion.
Complete cosmological target	NotReady _{JLR-D4-1}	No fresh complete fibre or base-image theorem.
Nonstandard histories	ROOS or not ready	Authorized histories retained outside the standard summary response.
Astrophysical successor face	Not ready	No target-complete astrophysical packet.
Summary-target uniqueness	False	Continuum live residue.

17 Physical interpretation and cross-phenomenon comparison

17.1 What the summary target has selected

The positive D4 theorem selects neither one PCR-5B base class nor a proper subset. It restricts the ordering-resolved scalar mass-sum coordinate within every base fibre. The exact statement is about $T_{D4}^{\text{pub95-summary}}$, not about the full joint posterior geometry and not about T_{D4}^{complete} .

The inequality

$$\Sigma_{\text{NO}}(0) < \Sigma_{95}^{\text{pub}} < \Sigma_{\text{IO}}(0)$$

places the standard inverted-ordering mass-sum image outside the frozen scalar band. It does not govern partial population, decay, interaction, hidden radiation, alternative background, or other response-changing histories.

17.2 Why D4 is not merely D1 repeated

PCR-5D1 made absolute support scale standing. D4 embeds that scale into a source-authorized thermal, background, and transport envelope and then applies a cosmological scalar summary rule. Thus the broader ledger map is

$$\text{AbsoluteScale} \mapsto \text{CosmologicalScaleHistory}.$$

On the fixed positive branch, however, the exact minimal separator actually proved is Σ_{ν} . The fixed thermal/free-streaming context is physically necessary for the realization but has not been independently varied to prove minimal standing separation.

17.3 Why D4 is not the D3 selector

D3 target 1.3 eliminates a proper continuum subset of the oscillation base through retained active-process closure. D4 has full base image at the summary target and acts through a scalar mass-sum refinement. Their images and minimal separators are therefore distinct. A common generator remains a PCR-5E question requiring a factorization theorem on one common physical point.

17.4 Seven-coordinate interpretation

$$\tau_{\text{cos}} = (A, W, Q, C, B, E, M)$$

keeps the background anchor, population/transport witnesses, summary or observable quotient, continuation, calibration boundary, estimator rule, and hidden remainder distinct. This follows the formal-versus-executable separation of the companion transport-closure analysis. In PCR-5D4 the seven-coordinate record is downstream: it audits a physically formed point and cannot create a source, repair a failed cross relation, or convert a scalar summary into a complete likelihood.

Structural significance

Cosmology has sharpened every oscillation-carrier fibre without selecting one base carrier. The standard summary branch leaves a continuous normal-ordering mass-sum region over every base class. The complete cosmological response remains unresolved, so the result is strong but deliberately target-relative.

18 Empirical and computational certificate record

18.1 Primary CMB initialization

The production CMB packet was initialized with Cobaya 3.6.2 and CAMB 1.6.6. The selected components were Planck low- ℓ TT and EE, NPIPE CamSpec TTTEEE, and ACT DR6 plus Planck lensing. The package audit inventoried 1,782 files with an SHA-256 record for each. The Planck-only fixed point reproduced the reference values to below 7.4×10^{-4} absolute χ^2 .

Table 5: Fixed-point primary CMB initialization.

Component	Planck-only χ^2	Combined-stack χ^2
Planck low- ℓ TT	24.809301	24.809298
Planck low- ℓ EE	395.719300	395.719300
NPIPE CamSpec TTTEEE	11337.339271	11334.006658
ACT DR6 plus Planck lensing	–	21.492304

The combined CamSpec value differs from the isolated value because the lensing likelihood requests CMB-dependent correction terms and therefore changes the common theory-evaluation context. One auxiliary standalone ACT diagnostic passed an obsolete parameter at the wrong configuration level; it is retained as a failed diagnostic and does not override the successful combined target construction.

18.2 Joint DESI released scalar summary record

The selected released branch contains four chains with 348,099 saved rows, total stored weight 428,587, and weight-based effective sample size 291,843.86. Classic chain diagnostics are all close to unity:

$$\hat{R}(m_\nu) = 1.000416, \quad \hat{R}(H_0) = 1.000213, \quad \hat{R}(\Omega_m) = 1.000309, \quad \hat{R}(\sigma_8) = 1.000614.$$

The exact released m_ν marginal used by the summary-target theorem is summarized in Table 6 and Figure 4.

Table 6: Weighted released m_ν marginal, in eV.

Quantile	2.5%	5%	16%	50%
Value	0.000798	0.001567	0.005271	0.018346
Quantile	84%	90%	95%	97.5%
Value	0.041599	0.050190	0.061935	0.072987

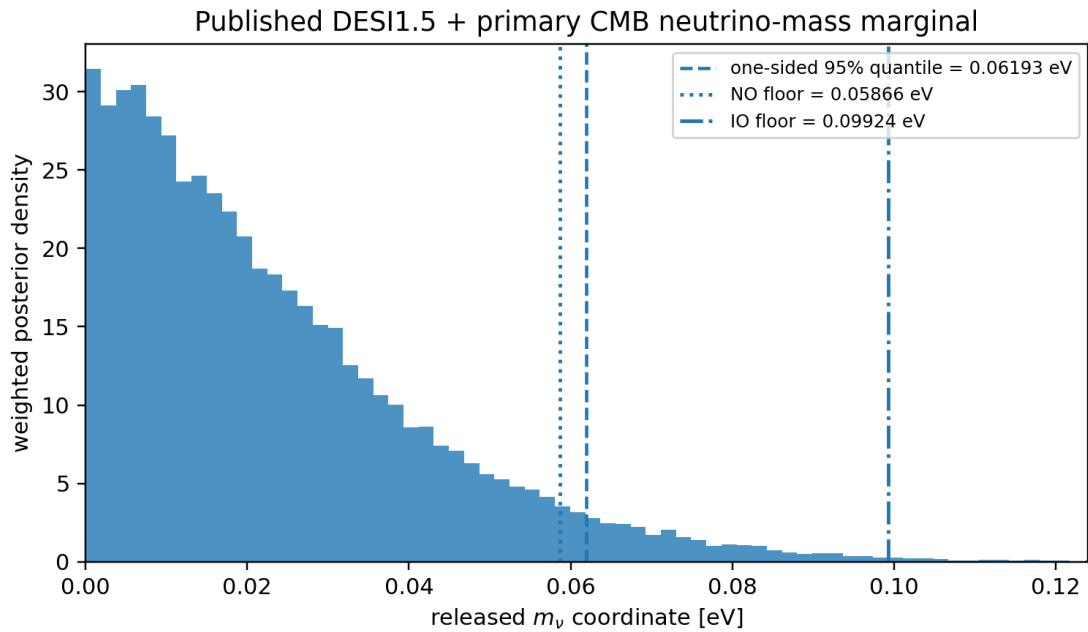


Figure 4: Exact weighted marginal from the retained released scalar record ensemble. The vertical lines show the normal-ordering floor, released one-sided 95% quantile, and inverted-ordering floor used in the summary-target theorem.

18.3 Mass-sum branch geometry

The rounded oscillation projection follows the standard three-neutrino ordering convention and the current global-fit scale of the measured splittings [10]. Figure 5 shows the two strictly increasing mass-sum maps and their relation to the released quantile.

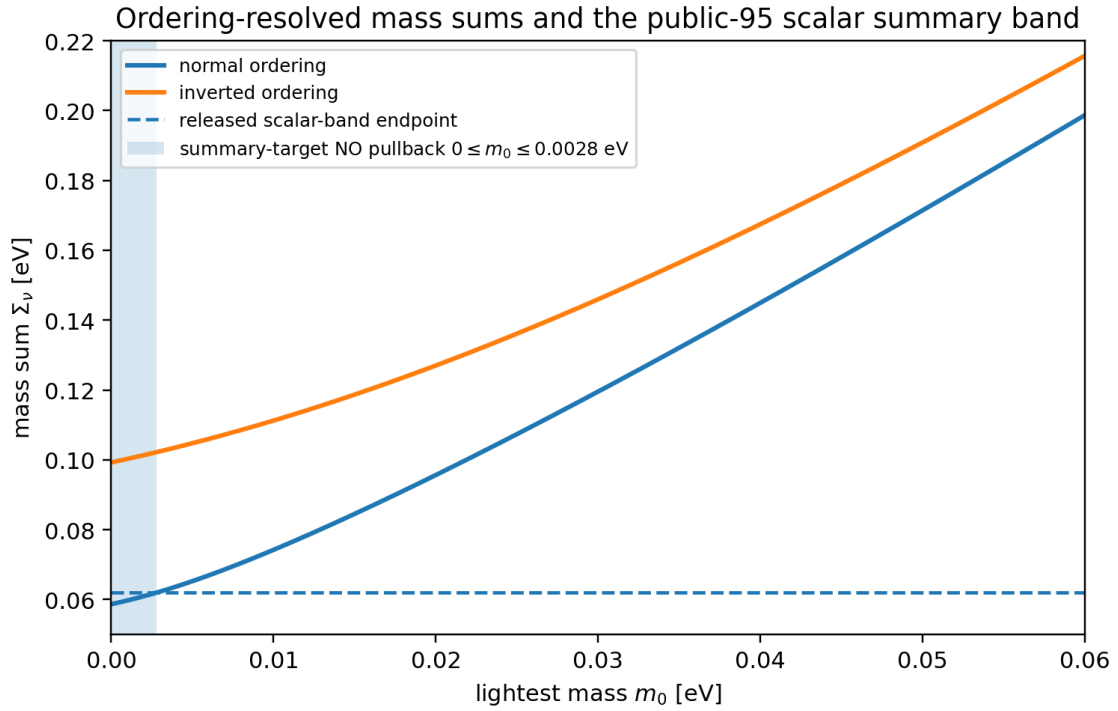


Figure 5: Normal- and inverted-ordering mass-sum maps under the frozen oscillation projection. The normal branch intersects the released 95% line at m_* ; the inverted branch begins above it.

18.4 Released covariance initialization

The public joint release contains sixteen 6×6 covariance objects. Every entry is finite. The raw matrices contain small finite-chain antisymmetric remainders; the frozen canonical rule is

$$C_{\text{sym}} = \frac{C + C^T}{2}.$$

After this declared normalization all sixteen are positive definite and admit Cholesky factorization. Across the set,

$$\max \|C - C^T\|_{\infty} = 5.0981 \times 10^{-6},$$

$$\min \lambda_{\min}(C_{\text{sym}}) = 2.2167 \times 10^{-5}, \quad \max \kappa(C_{\text{sym}}) = 609.9004.$$

This is structural covariance initialization. It does not supply the missing fresh theory-to-likelihood map.

18.5 Computational authority boundary

The numerical records certify exact bytes, hashes, chain ingestion, weighted summaries, covariance normalization, and fixed-point likelihood construction. They do not

prove the semantic completeness of SOE-D4-1, the physical-apex theorem, the retained-transparency theorem, or the groupoid fibre theorems. Conversely, the semantic theorems do not recompute a collaboration posterior. The scientific result requires the two layers to meet only at the explicitly declared certificate interfaces.

19 Hostile-referee closure map

Table 7: Principal hostile attacks, manuscript closure, and faithful reopening condition.

Attack	Present closure	Faithful reopening condition
Target laundering	Theorem 3.4 freezes the summary target independently and proves it non-equivalent to the complete target.	Show that the scalar band or rule was chosen using the carrier endpoint, or prove a target equivalence to T_{D4}^{complete} .
Posterior-as-likelihood substitution	The summary target uses only set membership in B_{95}^{pub} ; JLR remains negative.	Positive JLR Route A or B.
Scalar release coordinate treated as ordering posterior	Theorem 7.2 permits only a set-theoretic pullback and explicitly denies measure transport.	Supply an exact mass-prescription and posterior-measure transport theorem.
Source-law suppression	KSD-D4-1 assigns every old active relation a D4 landing value; Lemma 5.4 nulls only the complement of the saturated support.	Exhibit a mandatory kinetic or stress landing omitted from the complete trace; this is a fixed-scope counterexample or source update.
Mass-dependent background reuse	BG-D4-1 constructs $\mathcal{B}_{\omega, m_0}$, preserves the present matter anchor by a deterministic CDM compensation, and proves positivity on the full NO interval.	Exhibit a mass point in the interval for which the compensation is non-physical or the Friedmann/perturbation initial-value problem fails.
Kinetic consequence suppressed by projection	KSD-D4-1 closes the complete target seed under every explicit source-active production, collision, decay/injection, environmental, constituent/collective, and stress-energy relation.	Exhibit an active typed D4 forcing edge omitted from $\mathcal{R}_{\text{kin}}^{D4}$.
Seed-active coherence failure	SAR-D4-1 audits the frozen SOE-D4-1 activity-support schemas and proves the seed-active/off-seed-null Boolean vector is a coherent source point; activity maps are primitive source parameters/guards and no frozen clause derives one from another.	Exhibit a frozen v1.0.0 constitutive clause with activity-support consequent, or a cross-activity implication forcing an off-seed map true. This defeats SAR-D4-1 and reopens the downstream section.
Hidden or hybrid source called null because the response omits it	SAR-D4-1 preserves the actual old HID/HYB data while proving only the new off-seed D4 activity maps are source-coherently false; STR-D4-1 then evaluates their kinetic landing with KSD-D4-1.	Exhibit a faithful same-scope source law forcing independent off-seed D4 activity; this is a genuine Source-Update or nontransparent/base-selection result.
Stress energy omitted or double counted	KSD-D4-1 forms the full stress diagram and takes its colimit before descent to the three compiled modes.	Exhibit a source-authorized nonzero stress contribution outside the diagram or an uncoequalized duplicate contribution.
Component splicing	D4-03 constructs one combined semantic object only after KSD-D4-1 and BG-D4-1 discharge source, stress-energy, and background closure.	Exhibit a declared cross relation or stratum with no common apex.
Missing stratum	STR-D4-1 treats LAW, LOCAL, HID, COMP, SOL, and HYB at the actual seeded activity assignment.	Exhibit an authorized PCR-5B stratum outside the table or a source equation not discharged at the seeded point.

Attack	Present closure	Faithful reopening condition
Deletion-based transparency	D4-04 defines the D4-only ideal and proves channelwise equality with all D4 fields present.	Exhibit an admitted D4-to-old-output constructor or changed retained channel.
Base merger	D4-04B restricts enriched arrows to base arrows.	Supply a same-scope enriched arrow connecting distinct PCR-5B components.
Boolean threshold collapse	SSA-D4-1 fixes S_Σ as an exact standing output and factors the Boolean rule through it.	Prove that the target retains only Acc_{95} and lawfully forgets S_Σ .
Finite sample counting	D4-05 uses the analytic interval, strict monotonicity, and scalar standing activation.	Supply a lawful standing identification collapsing unequal S_Σ values.
IO physical exclusion	D4-11 assigns summary out-of-band, not solved empty.	Complete evaluator at the same target showing a different disposition.
Standard summary applied to nonstandard history	D4-11 returns ROOS or NotReady.	History-specific response theorem and target packet.
False minimal separator	D4-09 names Σ_ν as the only minimal proved separator and retains scale history as a ledger class.	Evaluate a variable-history branch and prove a smaller or additional minimal separator.
Macroblock profile promoted to exact PCR-5C profile	D4-07 labels the result $Y_{\text{compiled},3\text{-block}}$ and retains the fine deletion hypergraph as readiness-gated.	Supply the base-dependent constitutively closed deletion calculation on every compiled stratum.
Response counted as source	D4-07 separates the declared source macroblocks from complete-realization dependence.	Prove that a target response block is an occupancy-bearing source constituent under the frozen source theory.
D3 common-generator inference	D4-12 exports DistinctSeparator only.	PCR-5E factorization through one retained invariant.
Summary leg substituted for complete D4 leg	D4-12 exports two typed legs.	Positive JLR status or a proved target equivalence.

20 Reopening ledger

Table 8: Exact conditions that reopen a D4 theorem.

New fact	Status effect	Reopened result
A participating history requires a new source sort, operation, or law.	SourceUpdate	D4-02 and all downstream history claims.
A frozen or newly proved same-scope source clause forces an off-seed D4 activity map from the active seed.	SourceUpdate	SAR-D4-1, STR-D4-1, D4-03 and every downstream fibre/image theorem.
An explicit source-active D4 kinetic or stress-energy relation forces support beyond the retained interface.	Fixed-source non-transparent branch	KSD-D4-1, D4-03, and the full-base theorem for the affected source history; the canonical representative remains unchanged only if its complete trace lacks that relation.
A new theorem derives a mandatory D4 production or stress map from an old-only coupling.	SourceUpdate	Typed non-generation, KSD-D4-1, and every downstream physical/fibre theorem.
A compiled stratum has no common physical apex.	Fixed-target counterexample	D4-03 through the capstone.
A lawful compiled history changes a retained oscillation channel in situ.	Transparency counterexample	D4-04 and every quotient/fibre result.
The public scalar record or summary functional changes.	TargetUpdate	Summary target identity and every summary-dependent theorem.
A publisher joint likelihood becomes available and passes Route A.	JLR Route A	Complete-target fibres and base image become evaluable.
An independent reconstruction passes Route B.	JLR Route B	Same complete-target theorems.

New fact	Status effect	Reopened result
A reconstruction fails identity, covariance, response, prior, derivative, or mutation checks.	NotReady	Complete target remains blocked; summary theorem unchanged.
A certified approximation changes semantic history or summary standing.	Standing split or Undef	D4-06 and D4-10.
A target-complete astrophysical packet is frozen.	Successor target	D4-01 through D4-12 on the new branch.
A D3/D4 factorization through one invariant is proved.	PCR-5E candidate	D4 membership unchanged; comparison tag may be refined.

21 Conclusion

PCR-5D4 now has an exact two-level endpoint. The positive target is the explicitly frozen public-95 scalar summary target $T_{D4}^{\text{pub95-summary}}$. Its candidate histories are source-first; SAR-D4-1 first proves the actual seed-active/off-seed-null activity vector is a coherent relative extension of the frozen source theory, STR-D4-1 applies the KSD dichotomy on that coherent point, and KSD-D4-1 then supplies the exact source-faithful kinetic closure, off-support kernel zeros, and stress-energy descent before the one-point lift is formed. The complete retained oscillation packet is unchanged in situ.

The released scalar band has the exact ordering-resolved pullbacks

$$\Sigma_{\text{NO}}^{-1}(B_{95}^{\text{pub}}) = [0, m_{\star}], \quad \Sigma_{\text{NO}}(m_{\star}) = \Sigma_{95}^{\text{pub}}, \quad m_{\star} \approx 0.002766609601185372 \text{ eV},$$

and

$$\Sigma_{\text{IO}}^{-1}(B_{95}^{\text{pub}}) = \emptyset.$$

These are set-theoretic summary-band statements, not ordering-conditioned posterior intervals.

For every $q \in Q_{\text{osc}}^{\text{SCIC}}$, the normal-ordering interval gives continuum many standing-distinct summary-target lifts. Hence

$$\boxed{\begin{aligned} \text{Status}(D4; T_{D4}^{\text{pub95-summary}}) &= \text{FullBaseContinuum}_{D4}^{\text{pub95-summary}}, \\ \text{im } \rho_{D4}^{\text{pub95-summary}} &= Q_{\text{osc}}^{\text{SCIC}}, \\ |\pi_0 \mathcal{L}_{D4}^{\text{pub95-summary}}(q)| &\geq \mathfrak{c} \quad (q \in Q_{\text{osc}}^{\text{SCIC}}). \end{aligned}}$$

The minimal new separator on the fixed branch is Σ_{ν} , whose exact value is target-activated by S_{Σ} . The broader cosmological scale-history object remains the physical response ledger. The proved source profile is $Y_{\text{compiled,3-block}}$; the fine PCR-5C deletion hypergraph remains explicitly **NotReady**_{FineSourceDeletion}, while background and response are recorded separately as complete-realization dependencies.

The complete cosmological target is not repaired by the summary theorem:

$$\boxed{\text{Status}(D4; T_{D4}^{\text{complete}}) = \text{NotReady}_{\text{JLR-D4-1}}.}$$

A positive JLR route would reopen the complete-target fibres and image; until then, non-standard histories and fresh arbitrary-theory calculations remain fail-closed. This preserves the strongest defensible result without assigning theorem authority to a missing likelihood or turning the project into an executable formalization programme.

The resulting D4 handoff to PCR-5E therefore contains two explicitly tagged marginal legs: a positive full-base continuum summary leg and a complete-target **NotReady** leg. PCR-5E must preserve that distinction when it forms the simultaneous joint target.

A Guarded standard-thermal compiler normal form

For each framed base source H , ordering ω , and lightest mass m_0 , the normal form is

$$\text{CTH}_\omega(H, m_0) = (H, \text{P}_{\text{osc}}(H), a_H^{D4}, M_\omega(m_0), f^{\text{FD}}, \eta_{\text{dec}}, g_{\Lambda\text{CDM}}, T_\nu^{\mu\nu}, \mathcal{B}, \Pi, \mathfrak{A}_0).$$

It is formed in the order:

1. retain the full old source and physical base;
2. identify the target-visible propagation seed;
3. apply SAR-D4-1 to construct the seed-active relative extension and prove the complete frozen activity/coherence subsystem satisfied;
4. restrict that relative extension through STR-D4-1 to the chosen base representative, with each off-seed D4 relation carrying its source-coherently false activity-support map;
5. close that seed at the coherent STR-D4-1 source point under every explicit source-active D4 kinetic and stress-energy relation to form KinSat_{D4} ;
6. apply guarded null participation only on the complement of KinSat_{D4} ;
7. attach ordering-resolved masses while preserving the splittings;
8. attach the standard distribution and history;
9. form stress energy and background coupling;
10. evolve the perturbation hierarchy;
11. form response and summary-certificate data;
12. evaluate target status last.

Target data enter only at the last two arrows. No scalar band is used to generate a source history or physical apex.

A.1 Stratumwise kinetic and physical packet

The LAW, LOCAL, HID, COMP, SOL, and HYB cases use the complete kinetic-saturation trace recorded in Theorem 6.3 and the explicit partially active source points constructed by Theorem 6.5. In each case the old source-active packet is retained; hidden coupling, threshold, memory, constituent, collective, gluing, and higher-incidence data are inspected; D4 participation is null only outside the least source-active kinetic closure; and the full stress-energy diagram is coequalized before its exact three-mode descent.

A.2 Retained-base transparency packet

The D4-only ideal is

$$\mathcal{I}_{D4}^{\text{th}} = \langle \text{Pop}, \text{ThermalHistory}, \text{BgRef}, \text{StressEnergy}, \\ \text{Perturb}, \text{Response}, \text{Approx}, \text{D4-only incidence} \rangle.$$

The compiler contains no generator from this ideal to an old oscillation-output sort. The transparent packet therefore requires and proves the channelwise equalities of Theorem 6.8. Equality after deleting D4 data is insufficient.

B Exact public scalar-band mass-sum certificate

The retained numerical record supplies the scalar band endpoint and chain provenance. It does not define an ordering-resolved posterior. The mathematical certificate consists of:

$$\begin{aligned} B_{95}^{\text{pub}} &= (-\infty, 0.061934524 \text{ eV}], \\ \Sigma_{\text{NO}}(m_0) &= m_0 + \sqrt{m_0^2 + \delta} + \sqrt{m_0^2 + \Delta}, \\ \Sigma_{\text{IO}}(m_0) &= m_0 + \sqrt{m_0^2 + \Delta - \delta} + \sqrt{m_0^2 + \Delta}. \end{aligned}$$

B.1 Monotonicity and uniqueness

Both maps have strictly positive derivative. Since

$$\Sigma_{\text{NO}}(0) < 0.061934524 < \Sigma_{\text{IO}}(0),$$

there is a unique normal-ordering endpoint and no inverted-ordering preimage.

B.2 Exact numerical ledger

Quantity	Value
Released scalar 95% endpoint	0.061934524 eV
NO floor	0.0586602540 eV
IO floor	0.0992442890 eV
NO lightest-mass endpoint	unique root of $\Sigma_{\text{NO}}(m_*) = \Sigma_{95}^{\text{pub}}$, with $m_* \approx$ 0.002766609601185372 eV
NO summary-compatible set	$[0, m_*]$
IO summary-compatible set	$\emptyset$

The certificate explicitly denies posterior-measure transport, ordering-conditioned credible-interval semantics, and fresh likelihood authority.

C Primary observational packet and covariance certificate

C.1 Primary CMB byte identity

The complete file-level SHA-256 ledger is retained in

reference_records/CMB_PRIMARY_AUDIT_STATUS.json

and in the upstream payload-completion project. The present manuscript packages the status record rather than redistributing multi-gigabyte collaboration likelihood payloads.

C.2 Joint DESI public archive

The retained publisher archive has:

bytes = 305,659,195,

MD5 = ff5e5ec8c844a36ffa30b93216a5c740,

SHA256 = 6fc46986035bd6a63e8d2011e56ee384 b1235381f5a5581962860e1f59023c43.

The exact posterior chain and covariance hashes are included in the structured records. Archive identity and posterior initialization do not close JLR-D4-1.

D JLR-D4-1 validation specification

D.1 Route A record

A Route A certificate must include:

- publisher, repository, tag/commit, release, license, and byte identity;
- installation and initialization transcript;
- observable vector ordering, units, masks, and covariance construction;
- nuisance, calibration, prior, and approximation ledger;
- deterministic reference points and expected log-likelihood values;
- derivative and parameter-transport checks;
- released-posterior reproduction report;
- mutation results and package manifest.

D.2 Route B record

A Route B certificate must additionally document every independently reconstructed equation and its source correspondence. Agreement only at one posterior location is insufficient. The reconstructed evaluator must be total on the declared domain and must reproduce both reference values and the relevant posterior geometry under the frozen priors and nuisance treatment.

D.3 Mandatory mutations

1. permute one ShapeFit/BAO observable block;
2. change one unit conversion;
3. remove one off-diagonal covariance block;
4. replace C_{sym} by the raw matrix without declaration;
5. delete one nuisance or calibration parameter;
6. shift one prior boundary;
7. alter one mask or scale cut;
8. change the nonlinear or bias prescription;
9. replace the evaluator by posterior interpolation;
10. tune one reconstruction choice after observing the D4 endpoint.

Each mutation must change or invalidate the appropriate reference output. A silent pass is a fidelity failure.

E Status and failure-certificate schema

The machine-readable status record for each raw point should contain at least:

```
{
  "base_class": "...",
  "source_id": "...",
  "history_branch": "...",
  "target_branch": "...",
  "physical_apex": "pass|fail|not_ready",
  "retained_transparency": "pass|fail|not_ready",
  "response_applicable": true,
  "approximation_certificate": "...",
  "summary_interface": "in_band|out_of_band|not_applicable",
  "complete_interface": "available|missing|out_of_scope",
  "status": "...",
  "failed_constraint": "...",
  "provenance": ["..."]
}
```

The validator must reject records that call an ill-typed, ROOS, published-out, or JLR-blocked point SolvedEmpty.

F Machine-readable theorem and handoff records

The project contains the following authoritative structured outputs:

- PCR5D4_THEOREM_CHAIN_v1.5.0.json;
- PCR5D4_SUMMARY_TARGET_FREEZE_v1.5.0.json;
- PCR5D4_TARGET_AND_STATUS_v1.5.0.json;
- PCR5D4_KINETIC_SATURATION_AND_STRESS_ENERGY_v1.5.0.json;
- PCR5D4_SEEDED_TRANSPARENT_SECTION_v1.5.0.json;
- PCR5D4_PHYSICAL_AMALGAMATION_LEDGER_v1.5.0.json;
- PCR5D4_TRANSPARENCY_LEDGER_v1.5.0.json;
- PCR5D4_FIBRE_DISPOSITION_LEDGER_v1.5.0.json;
- PCR5D4_STANDING_COORDINATE_LEDGER_v1.5.0.json;
- PCR5D4_SOURCE_AND_REALIZATION_DEPENDENCY_v1.5.0.json;
- PCR5D4_JLR_GATE_v1.5.0.json;
- PCR5D4_PCR5E_HANDOFF_v1.5.0.json;
- PUBLIC_SCALAR_SUMMARY_BAND_CERTIFICATE_v1.5.0.json, containing released-record provenance and the probability-authority boundary;
- PUBLIC_SCALAR_BAND_MASS_SUM_CERTIFICATE_v1.5.0.json, containing the deterministic ordering-resolved set-theoretic pullback.

The JSON records are audit and transport surfaces. The mathematical theorem authority remains the manuscript proof chain and its explicitly imported results.

G Notation summary

Table 9: Principal notation.

Symbol	Meaning
$Q_{\text{osc}}^{\text{SCIC}}$	Exact SCIC connected-component quotient of the frozen oscillation target.
$\mathcal{N}(T_{\text{osc}})$	Oscillation standing-core groupoid.
$\mathfrak{U}_{\nu, D4}^{\text{src}}$	SOE-D4-1 source universe of participating cosmological neutrino histories.
$T_{D4}^{\text{public release}}$	Exact retained public-release record.
$T_{D4}^{\text{pub95-summary}}$	Frozen public-95 scalar summary target on the standard thermal branch.
T_{D4}^{complete}	Complete cosmological target gated by JLR–D4–1.
$\mathcal{N}_{D4}^{\text{pub95-summary}}$	Public-95 summary-target D4 standing groupoid.
$\rho_{D4}^{\text{pub95-summary}}$	Connected-component restriction map to $Q_{\text{osc}}^{\text{SCIC}}$.
$\mathcal{L}_{D4}^{\text{pub95-summary}}(q)$	Connected-base D4 lift fibre over q .
ω	Mass ordering, NO or IO.
m_0	Lightest mass in the ordering-adapted parameterization.
$\Sigma_{\omega}(m_0)$	Ordering-specific neutrino mass sum.
$m_{\star}$	Unique normal-ordering endpoint of the scalar-band pull-back.
τ_{cos}	Seven-coordinate inference certificate (A, W, Q, C, B, E, M) .
JLR–D4–1	Named fresh joint-DESI likelihood execution gate.
FullBaseContinuum	Full base image with continuum summary-target fibre in every base component.

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